Heat Kernels and Index Theory

Based on lectures by Paul Loya (2013)

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Chapter 1

Geometric preliminaries

Throughout, M is a smooth compact manifold without boundary unless stated otherwise. All vector bundles are finite-dimensional and smooth.

Discussion (Scope of the proofs). These notes are proof-oriented, but the following results are imported, with citations, and are not proved here.

1. *Topological K-theory*: Bott periodicity, the Thom isomorphism, and functoriality and excision of the K-theoretic push-forward [1, 3].
2. *The analytic push-forward theorem* (Theorem 2.19) for embeddings of closed manifolds [1].
3. *From K-theory to cohomology*: the cohomological form of the topological index obtained from the Chern character and the cohomological Thom isomorphism (Theorem 2.17), and the fact that, rationally, symbols of twisted Dirac (indeed twisted signature) operators generate $K_c^0(T^*M)$, so that the Dirac case of the cohomological formula implies the general case [2, 6], [4, Chapter 4].
4. *The rescaled heat-kernel theorem* for Dirac operators, including Getzler's filtration bound on the heat coefficients [4, Chapters 4–5], [5].
5. *The blockwise scalar Mehler formula* [4, Chapter 4].
6. *The Newlander–Nirenberg theorem*, used only to name integrable almost complex structures.

The arguments below prove the remaining reductions and indicate where each imported result enters.

1.1 Connections and curvature

Definition 1.1. A connection on a real or complex vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \longrightarrow \Omega^1(M; E)$$

satisfying

$$\nabla(fs) = df \otimes s + f\nabla s.$$

For a vector field X, write $\nabla_X s = (\nabla s)(X)$.

In a local frame $e = (e_1, \dots, e_r)$, a section is a column vector and

$$\nabla = d + \omega, \qquad \omega \in \Omega^1(U; \operatorname{End} E).$$

If the frame changes by $e' = eg$, then

$$\omega' = g^{-1}\omega g + g^{-1}dg.$$

This transformation law is exactly what makes the definition independent of a frame.

The connection extends to E-valued forms by

$$d_\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^{\deg \alpha}\alpha \wedge \nabla s.$$

Definition 1.2. The curvature is

$$F^\nabla = d_\nabla^2 \in \Omega^2(M; \operatorname{End} E).$$

In a local frame,

$$F^\nabla = d\omega + \omega \wedge \omega.$$

Because d_∇^2 is $C^\infty(M)$-linear, it is an endomorphism-valued two-form. The graded Jacobi identity gives the Bianchi identity

$$d_\nabla F^\nabla = 0.$$

Question 1.3. *Why is a connection not tensorial while its curvature is?*

Answer. The Leibniz rule contains the derivative df, so $\nabla(fs) \neq f\nabla s$. Applying the connection twice makes the first-derivative terms cancel:

$$d_\nabla^2(fs) = d_\nabla(df \otimes s + f\nabla s) = -df \wedge \nabla s + df \wedge \nabla s + f\, d_\nabla^2 s.$$

Thus $d_\nabla^2(fs) = f\, d_\nabla^2 s$. This calculation is the intrinsic reason the inhomogeneous term $g^{-1}dg$ in the transformation law for ω disappears from the transformation law $F' = g^{-1}Fg$. □

If E is Hermitian, a connection is unitary when

$$X\langle s_1, s_2\rangle = \langle \nabla_X s_1, s_2\rangle + \langle s_1, \nabla_X s_2\rangle.$$

In a unitary frame, its connection form is skew-Hermitian and its curvature is skew-Hermitian.

1.2 The Levi–Civita connection

Let g be a Riemannian metric. A connection on TM is metric if $\nabla g = 0$, and its torsion is

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

Theorem 1.4 (Levi–Civita)**.** *There is a unique metric connection on TM with zero torsion. It is determined by the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ & - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]). \end{aligned}$$

Proof. Metric compatibility expresses $Xg(Y,Z)$, $Yg(Z,X)$, and $Zg(X,Y)$ through covariant derivatives. Add the first two expressions, subtract the third, and use zero torsion to obtain the displayed formula. It proves uniqueness.

For existence, denote the right-hand side of the Koszul formula by $K(X,Y,Z)$. The derivatives of a function multiplying Z cancel, so $K(X,Y,fZ) = fK(X,Y,Z)$. Nondegeneracy of g therefore defines a unique vector field $\nabla_X Y$ by $2g(\nabla_X Y, Z) = K(X,Y,Z)$. Expanding the same formula gives

$$K(fX,Y,Z) = fK(X,Y,Z), \qquad K(X,fY,Z) = fK(X,Y,Z) + 2X(f)g(Y,Z).$$

Hence $\nabla_{fX}Y = f\nabla_X Y$ and $\nabla_X(fY) = X(f)Y + f\nabla_X Y$, so this is a connection. Finally,

$$K(X,Y,Z) - K(Y,X,Z) = 2g([X,Y],Z)$$

proves that its torsion vanishes, while

$$K(X,Y,Z) + K(X,Z,Y) = 2Xg(Y,Z)$$

proves metric compatibility. Thus the connection defined by the formula has both required properties. □

In local coordinates $x^1,\dots,x^n$, write $\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k \partial_k$. Substitution in the Koszul formula gives

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}\left(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\right).$$

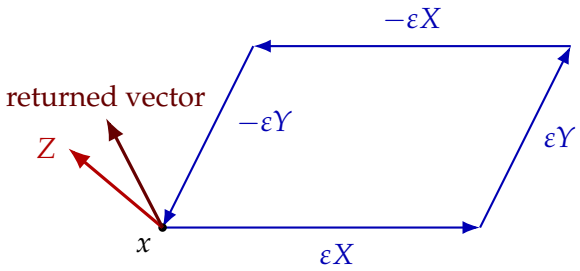
In geodesic normal coordinates centred at p, radial lines are geodesics, so $\Gamma_{ij}^k(p)x^i x^j = 0$ for all x; by symmetry of Γ_{ij}^k in i,j, $\Gamma_{ij}^k(p) = 0$, while $g_{ij}(p) = \delta_{ij}$ because $d\exp_p$ is the identity at the origin.

The Riemann curvature tensor is

$$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

In the same convention,

$$R(\partial_i,\partial_j)\partial_k = R^\ell{}_{kij}\partial_\ell, \qquad R^\ell{}_{kij} = \partial_i\Gamma_{jk}^\ell - \partial_j\Gamma_{ik}^\ell + \Gamma_{jk}^m\Gamma_{im}^\ell - \Gamma_{ik}^m\Gamma_{jm}^\ell.$$



$$\text{returned vector} = Z - \varepsilon^2 R(X,Y)Z + O(\varepsilon^3)$$

Figure 1.1: Curvature is the second-order failure of parallel transport around an infinitesimal parallelogram to return a vector unchanged. Transporting Z around the loop with sides εX, εY, $-\varepsilon X$, $-\varepsilon Y$ (commuting X, Y) multiplies it by $I - \varepsilon^2 R(X,Y) + O(\varepsilon^3)$; in a local frame this is $\exp(-\oint \omega)$ to second order.

With $R(X,Y,Z,W) = g(R(X,Y)Z,W)$, it satisfies

$$\begin{aligned} R(X,Y,Z,W) &= -R(Y,X,Z,W),\\ R(X,Y,Z,W) &= -R(X,Y,W,Z),\\ R(X,Y,Z,W) &= R(Z,W,X,Y), \end{aligned}$$

together with the first Bianchi identity

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0.$$

The Ricci tensor and scalar curvature are the contractions

$$\mathrm{Ric}(X,Y) = \sum_j g(R(e_j,X)Y,e_j), \qquad \mathrm{scal} = \sum_j \mathrm{Ric}(e_j,e_j).$$

Index convention. Throughout, in an orthonormal frame $e_1,\dots,e_n$,

$$R_{ijk\ell} = R(e_i,e_j,e_k,e_\ell) = g\big(R(e_i,e_j)e_k,e_\ell\big). \tag{1.1}$$

Thus the sectional curvature of the plane spanned by e_i, e_j is $K(e_i,e_j) = R_{ijji}$, and $\mathrm{scal} = \sum_{i,j} R_{ijji}$; for the unit sphere, $R(X,Y)Z = g(Y,Z)X - g(X,Z)Y$ and $R_{ijji} = 1$ for $i \neq j$. The curvature matrix R^{TM} of the Levi–Civita connection in this frame has entries $(R^{TM})_{\ell k} = g(R(\cdot,\cdot)e_k,e_\ell)$, so that $R(X,Y)e_k = \sum_\ell (R^{TM})_{\ell k}(X,Y)e_\ell$.

1.3 Chern–Weil forms

Let f be an invariant polynomial on a matrix Lie algebra. The form $f(F^\nabla)$ is closed, and its de Rham class does not depend on the connection.

Proposition 1.5 (Transgression)**.** *For a smooth family ∇^t with curvature F_t,*

$$\frac{d}{dt} f(F_t) = d\big(k\, f(\dot\nabla^t, F_t, \dots, F_t)\big)$$

when f is homogeneous of degree k.

Proof. The difference of two connections is tensorial; consequently $\alpha_t := \dot\nabla^t$ is a globally defined element of $\Omega^1(M; \mathrm{End}\, E)$. In a local frame $\nabla^t = d + \omega_t$, it is represented by $\alpha_t = \dot\omega_t$. Since

$$F_t = d\omega_t + \omega_t \wedge \omega_t,$$

differentiation gives

$$\dot F_t = d\alpha_t + \omega_t \wedge \alpha_t + \alpha_t \wedge \omega_t = d_{\nabla^t}\alpha_t.$$

Polarize f to a symmetric k-linear invariant form. Then

$$\frac{d}{dt} f(F_t,\dots,F_t) = k\, f(d_{\nabla^t}\alpha_t, F_t, \dots, F_t).$$

For endomorphism-valued forms, invariance of f implies the covariant Leibniz identity

$$d\, f(B_1,\dots,B_k) = \sum_j (-1)^{\deg B_1 + \cdots + \deg B_{j-1}} f(B_1,\dots,d_{\nabla^t}B_j,\dots,B_k);$$

the commutator terms cancel inside f. Apply this with $B_1 = \alpha_t$ and $B_2 = \cdots = B_k = F_t$. The Bianchi identity $d_{\nabla^t} F_t = 0$ kills every term except the first, proving

$$\frac{d}{dt} f(F_t) = d\big(k f(\alpha_t, F_t, \dots, F_t)\big).$$

Integrating from $t = 0$ to 1 shows explicitly that two connections produce cohomologous characteristic forms. □

Discussion. Chern–Weil theory is the bridge between analysis and topology used later. The heat kernel produces a particular curvature polynomial. Transgression shows that its integral on a closed manifold is independent of the connection used to compute it.

For a complex bundle E, the Chern character form is

$$\mathrm{ch}(E, \nabla) = \mathrm{Tr} \exp\left(\frac{iF^\nabla}{2\pi}\right).$$

For a real oriented bundle with curvature matrix R, define

$$\widehat{A}(E, \nabla) = \det{}^{1/2}\left(\frac{R/(4\pi i)}{\sinh(R/(4\pi i))}\right),$$

and

$$L(E, \nabla) = \det{}^{1/2}\left(\frac{R/(2\pi i)}{\tanh(R/(2\pi i))}\right).$$

These power-series expressions are interpreted through invariant polynomials. Their cohomology classes are denoted $\mathrm{ch}(E)$, $\widehat{A}(E)$, and $L(E)$.

The determinant notation is shorthand for traces of power series. If $f(z) = 1 + O(z)$, then

$$\det f(F) = \exp\big(\mathrm{Tr} \log f(F)\big), \qquad \det{}^{1/2} f(F) = \exp\left(\frac{1}{2} \mathrm{Tr} \log f(F)\right).$$

Although these are written as infinite series, only finitely many terms can contribute in any fixed differential-form degree. Conjugating the curvature does not change its traces, so every coefficient is an invariant polynomial and Chern–Weil theory applies.

Question 1.6. *Why do traces and determinants appear in the index density?*

Answer. Curvature is an endomorphism-valued two-form, whereas the index density must be an ordinary scalar form. The trace turns powers of curvature into scalar forms and is additive under direct sum. Exponentiating traces packages the multiplicative characteristic sequences, such as $\widehat{A}$ and L, that arise from independent curvature two-plane blocks in the heat model. □

For a complex bundle with Chern roots x_j, the Todd form is

$$\mathrm{Td}(E) = \prod_j \frac{x_j}{1 - e^{-x_j}}.$$

The splitting principle makes this a well-defined polynomial in the Chern classes.

Curvature roots, the Pfaffian, and the Euler form

Let $V \to M$ be a real oriented Euclidean bundle of rank $2m$ with a metric connection, and let R be its curvature matrix in an oriented orthonormal frame, with the matrix convention of (1.1). The *Pfaffian* of a skew-symmetric matrix A of even-degree forms is

$$\mathrm{Pf}(A) = \frac{1}{2^m m!} \sum_{\sigma \in S_{2m}} \mathrm{sgn}(\sigma) A_{\sigma(1)\sigma(2)} \cdots A_{\sigma(2m-1)\sigma(2m)},$$

so $\mathrm{Pf}\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right) = a$ and $\mathrm{Pf}(gAg^{-1}) = \mathrm{Pf}(A)$ for $g \in SO(2m)$. The *Euler form* is

$$e(V, \nabla) = \mathrm{Pf}\left(\frac{R}{2\pi}\right) \in \Omega^{2m}(M),$$

a closed form whose class $e(V)$ is independent of the connection by the transgression argument above.

Curvature roots. By the splitting principle we may compute invariant polynomials of R as if $V = L_1 \oplus \cdots \oplus L_m$ were an orthogonal sum of oriented two-plane bundles with oriented orthonormal frames e_{2j-1}, e_{2j} and connections preserving the sum. Give each L_j the complex structure $Je_{2j-1} = e_{2j}$. Then the block of $R/2\pi$ on L_j is $\left(\begin{smallmatrix} 0 & x_j \\ -x_j & 0 \end{smallmatrix}\right)$, and on the complex line (L_j, J) the curvature acts by $-2\pi i\, x_j$. Thus

$$x_j = c_1(L_j, J)$$

is the j-th *Chern root* of (V, J): the eigenvalue of $iR/2\pi$ on the complex line (L_j, J). As endomorphisms of the real bundle, $R/(2\pi i)$ and $iR/2\pi$ have eigenvalues $\pm x_1, \ldots, \pm x_m$. Consequently

$$\mathrm{Pf}\left(\frac{R}{2\pi}\right) = \prod_{j=1}^{m} x_j, \qquad \widehat{A}(V) = \prod_{j=1}^{m} \frac{x_j/2}{\sinh(x_j/2)}, \qquad L(V) = \prod_{j=1}^{m} \frac{x_j}{\tanh x_j},$$

the last two because the defining power series are even. For an oriented surface with Gaussian curvature K, the curvature matrix in an oriented orthonormal frame has $(1,2)$-entry $K\,dA$, so $x_1 = e(TM) = K\,dA/2\pi$. The Euler form is odd under reversal of orientation, and this is the only place where the orientation of the roots matters; all formulas below use this convention.

1.4 Differential complexes

A differential complex is a sequence

$$0 \longrightarrow \Gamma(E^0) \xrightarrow{D_0} \Gamma(E^1) \xrightarrow{D_1} \cdots \xrightarrow{D_{r-1}} \Gamma(E^r) \longrightarrow 0, \qquad D_{j+1} D_j = 0.$$

Its cohomology is

$$H^j(D) = \ker D_j / \operatorname{im} D_{j-1}.$$

When these spaces are finite-dimensional, its Euler characteristic is

$$\chi(D) = \sum_j (-1)^j \dim H^j(D).$$

The basic example is the de Rham complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \longrightarrow 0.$$

Its cohomology is the de Rham cohomology of M, and its Euler characteristic is the topological Euler characteristic $\chi(M)$.

Chapter 2

Elliptic operators and index

2.1 Differential operators and principal symbols

A differential operator $P : \Gamma(E) \to \Gamma(F)$ of order at most m has the local form

$$P = \sum_{|\alpha| \le m} a_\alpha(x) \partial_x^\alpha.$$

Its principal symbol is

$$\sigma_m(P)(x, \xi) = \sum_{|\alpha| = m} a_\alpha(x) (i\xi)^\alpha : E_x \longrightarrow F_x.$$

Equivalently,

$$\sigma_m(P)(x, df) s = \lim_{\tau \to \infty} \tau^{-m} e^{-i\tau f} P(e^{i\tau f} s)(x).$$

The latter formula proves coordinate invariance.

Discussion. The oscillatory test $e^{i\tau f} s$ isolates the part of P in which all m derivatives fall on the phase. Derivatives of the section, the coefficients, or lower-order terms carry fewer powers of τ and disappear after multiplication by τ^{-m}. This is why the principal symbol is intrinsic although the coefficients a_α are not.

The symbol is multiplicative:

$$\sigma_{m+\ell}(PQ) = \sigma_m(P) \sigma_\ell(Q).$$

If metrics and a density are fixed, the formal adjoint satisfies

$$\sigma_m(P^*)(x, \xi) = \sigma_m(P)(x, \xi)^*.$$

Formal adjoints and the Hodge star

Formal adjoints are used repeatedly below. Fix Hermitian metrics on E, F and a positive smooth density on M. There is a unique differential operator

$$P^* : C^\infty(M; F) \longrightarrow C^\infty(M; E)$$

such that

$$\langle Pu, v \rangle_{L^2} = \langle u, P^* v \rangle_{L^2}$$

for all smooth sections. In a coordinate chart, repeated integration by parts moves each ∂^α from u to the product of the coefficient, density, and v. A partition of unity patches the result; uniqueness follows from nondegeneracy of the L^2-pairing. Moving all m derivatives produces complex conjugation and the factor $(-1)^m$; with the convention $\sigma_m(P)(x,\xi) = \sum a_\alpha (i\xi)^\alpha$, these combine exactly to give the displayed adjoint-symbol identity.

If M is oriented, the Hodge star is the complex-linear map

$$* : \Lambda^k T^*M_{\mathbb{C}} \longrightarrow \Lambda^{n-k} T^*M_{\mathbb{C}}$$

characterized by

$$\alpha \wedge *\overline{\beta} = \langle \alpha, \beta \rangle \, d\,\mathrm{vol}\,.$$

It satisfies $*^2 = (-1)^{k(n-k)}$ on k-forms and

$$d^* = (-1)^{n(k+1)+1} * d * \quad \text{on } \Omega^k(M). \tag{2.1}$$

Consequently

$$\sigma_1(d)(x,\xi) = i\,\varepsilon(\xi), \qquad \sigma_1(d^*)(x,\xi) = -i\,\iota_{\xi^\sharp},$$

and the de Rham operator has symbol

$$\sigma_1(d+d^*)(x,\xi) = i\big(\varepsilon(\xi) - \iota_{\xi^\sharp}\big), \qquad \sigma_1(d+d^*)^2 = |\xi|^2 I.$$

Question 2.1. *Which parts of this construction require the metric and orientation?*

Answer. The exterior derivative needs neither. The metric and density define the formal adjoint; orientation identifies the density with a volume form and thereby defines the Hodge star. The Laplacian and harmonic representatives therefore depend on the chosen metric, whereas their cohomology classes and the resulting Euler characteristic do not. □

Definition 2.2. The operator P is elliptic if $\sigma_m(P)(x,\xi)$ is invertible for every $\xi \neq 0$.

For a first-order complex $(E^\bullet, D_\bullet)$, the symbol sequence at $(x,\xi) \in T^*M \setminus 0$ is

$$0 \longrightarrow E^0_x \xrightarrow{\sigma(D_0)(x,\xi)} E^1_x \longrightarrow \cdots \longrightarrow E^r_x \longrightarrow 0.$$

The complex is elliptic when this sequence is exact for every nonzero ξ.

Choose bundle metrics. The associated Laplacians are

$$\Delta_j = D_j^* D_j + D_{j-1} D_{j-1}^*.$$

Proposition 2.3. *A complex is elliptic if and only if every Δ_j is elliptic.*

Proof. At a nonzero covector, put $a_j = \sigma(D_j)$. Since $a_j a_{j-1} = 0$,

$$\sigma(\Delta_j) = a_j^* a_j + a_{j-1} a_{j-1}^*.$$

Its quadratic form is

$$\langle \sigma(\Delta_j) v, v \rangle = \|a_j v\|^2 + \|a_{j-1}^* v\|^2.$$

The kernel is therefore $\ker a_j \cap (\operatorname{im} a_{j-1})^\perp$, which vanishes exactly when $\ker a_j = \operatorname{im} a_{j-1}$. □

2.2 Pseudodifferential parametrices

A classical pseudodifferential operator of order m has, in local coordinates, an oscillatory representation

$$Pu(x) = (2\pi)^{-n} \int e^{i(x-y)\cdot\xi} p(x,\xi) u(y)\, dy\, d\xi,$$

modulo a smooth kernel, where

$$p(x,\xi) \sim p_m(x,\xi) + p_{m-1}(x,\xi) + \cdots$$

and p_{m-j} is homogeneous of degree $m-j$ for large ξ. The composition formula is

$$p\#q \sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha p\, D_x^\alpha q, \qquad D_x = \frac{1}{i}\partial_x.$$

Theorem 2.4 (Elliptic parametrix)**.** *If $P \in \Psi^m(M;E,F)$ is elliptic, there is $Q \in \Psi^{-m}(M;F,E)$ such that*

$$QP = I - R_L, \qquad PQ = I - R_R, \qquad R_L, R_R \in \Psi^{-\infty}.$$

Proof. Set $q_{-m} = p_m^{-1}$ outside a neighborhood of the zero section. Suppose $q_{-m}, \ldots, q_{-m-j+1}$ have been chosen. The component of degree $-j$ in $q\#p - I$ is

$$q_{-m-j} p_m + \sum_{\substack{k+\ell+|\alpha|=j \\ k<j}} \frac{1}{\alpha!} \partial_\xi^\alpha q_{-m-k}\, D_x^\alpha p_{m-\ell}.$$

Therefore set

$$q_{-m-j} = -\left(\sum_{\substack{k+\ell+|\alpha|=j \\ k<j}} \frac{1}{\alpha!} \partial_\xi^\alpha q_{-m-k}\, D_x^\alpha p_{m-\ell} \right) p_m^{-1}.$$

This removes the next homogeneous error. To globalize, choose coordinate charts and bundle trivializations, quantize the local asymptotic inverses, and insert a partition of unity on both sides of each local operator. The principal parts agree on overlaps because they are determined intrinsically by p_m^{-1}; hence the first patched error has order -1. Repeating the same correction in the global symbol sequence removes successively all homogeneous errors. Borel summation then gives a left parametrix Q_L and, by the analogous right recursion, a right parametrix Q_R:

$$Q_L P = I - R_L, \qquad PQ_R = I - R_R, \qquad R_L, R_R \in \Psi^{-\infty}.$$

The single operator

$$Q = Q_L + Q_R - Q_R P Q_L$$

is two-sided modulo smoothing: substituting the two identities shows that both $QP - I$ and $PQ - I$ contain a smoothing factor. This is the standard globalization and asymptotic-completeness argument of the classical pseudodifferential calculus. □

Let $H^s(M;E)$ denote the Sobolev space of order s. An operator in Ψ^m extends boundedly

$$P : H^s(M;E) \longrightarrow H^{s-m}(M;F).$$

A smoothing operator maps every Sobolev space continuously into every higher Sobolev space and is compact as an operator $H^s \to H^s$.

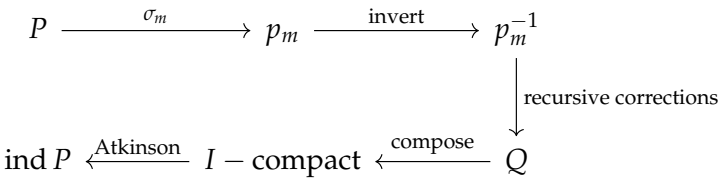


Figure 2.1: Ellipticity is converted into Fredholmness through symbol inversion, quantization, and compact remainders.

Theorem 2.5 (Elliptic regularity)**.** *If P is elliptic, $u \in \mathcal{D}'(M;E)$, and $Pu \in H^{s-m}(M;F)$, then $u \in H^s(M;E)$. More precisely, for $u \in H^s$,*

$$\|u\|_{H^s} \le C\big(\|Pu\|_{H^{s-m}} + \|u\|_{H^{s-1}}\big).$$

Proof. Every distribution on a compact manifold belongs to H^{-N} for some N. The parametrix identity, valid on distributions,

$$u = QPu + R_L u,$$

places the first term in H^s and the second in C^∞, proving regularity directly. For the estimate, use the mapping properties $Q : H^{s-m} \to H^s$ and $R_L : H^{s-1} \to H^s$:

$$\|u\|_{H^s} \le \|QPu\|_{H^s} + \|R_L u\|_{H^s} \le C\big(\|Pu\|_{H^{s-m}} + \|u\|_{H^{s-1}}\big).$$

This also records the lower-order term needed when the estimate is used in a compactness argument. □

Corollary 2.6. *On a compact manifold, an elliptic operator*

$$P : H^s(M;E) \longrightarrow H^{s-m}(M;F)$$

is Fredholm. Its kernel consists of smooth sections, and

$$\operatorname{coker} P \cong \ker P^*.$$

Proof. The parametrix identities make P invertible modulo compact operators, so Atkinson's theorem gives Fredholmness. Elliptic regularity makes the kernel smooth. The cokernel is the orthogonal complement of the range, which is $\ker P^*$. □

Theorem 2.7 (Hörmander–Fedosov trace formula)**.** *Let $Q \in \Psi^{-m}(M;F,E)$ be any parametrix for P and put*

$$R_E = I - QP, \qquad R_F = I - PQ.$$

Then both remainders are smoothing and

$$\operatorname{ind} P = \operatorname{Tr}(R_E) - \operatorname{Tr}(R_F).$$

Proof. Let Π_E and Π_F be the orthogonal projections onto $\ker P$ and $\ker P^*$. They have finite rank and smooth kernels. The generalized inverse $G : (\ker P^*)^\perp \to (\ker P)^\perp$, extended by zero on $\ker P^*$, satisfies

$$GP = I - \Pi_E, \qquad PG = I - \Pi_F.$$

The parametrix construction, followed by these finite-rank corrections, shows that $G - Q$ is smoothing. Put $K = G - Q$. Then

$$R_E = \Pi_E + KP, \qquad R_F = \Pi_F + PK.$$

On a closed manifold, a smoothing operator is trace class and its composition with a differential or pseudodifferential operator is again smoothing. Kernel composition and Fubini's theorem give $\operatorname{Tr}(KP) = \operatorname{Tr}(PK)$. Therefore

$$\operatorname{Tr}(R_E) - \operatorname{Tr}(R_F) = \operatorname{Tr}(\Pi_E) - \operatorname{Tr}(\Pi_F) = \dim\ker P - \dim\ker P^*.$$

□

Question 2.8. *Where are compactness of M and ellipticity used in Fredholmness?*

Answer. Ellipticity produces a parametrix whose remainders gain arbitrarily many derivatives. Compactness of M makes the Sobolev inclusion $H^{s+N} \hookrightarrow H^s$ compact, so those remainders are compact. On a noncompact manifold, translates of one bump function can remain bounded with no convergent subsequence; smoothing alone then need not imply compactness. The boundary calculus in the companion notes is designed to control exactly that failure. □

The analytic index is

$$\operatorname{ind} P = \dim\ker P - \dim\ker P^*.$$

It is independent of s, invariant under continuous elliptic deformations, and unchanged by lower-order perturbations.

2.3 Hodge theory

Hodge theory can be reached in two ways: from an elliptic parametrix and by integrating the heat semigroup. Both viewpoints are useful. The first proves closed range and finite-dimensionality; the second identifies the Green operator geometrically.

Let $(E^\bullet, D_\bullet)$ be an elliptic complex, so $D_j D_{j-1} = 0$, and put

$$\Delta_j = D_{j-1}D_{j-1}^* + D_j^* D_j, \qquad \mathcal{H}^j = \ker\Delta_j.$$

Lemma 2.9 (Energy identity)**.** *For every smooth section $u \in \Gamma(E^j)$,*

$$\langle \Delta_j u, u\rangle = \|D_{j-1}^* u\|^2 + \|D_j u\|^2.$$

Consequently

$$\mathcal{H}^j = \ker D_j \cap \ker D_{j-1}^*.$$

Proof. Move one factor in each summand of Δ_j across the inner product. Both terms are nonnegative, so their sum vanishes precisely when both operators annihilate u. □

The analytic input is the following estimate. It is the point at which ellipticity and compactness of the manifold enter the proof.

Proposition 2.10 (Elliptic estimate and compact resolvent)**.** *For every $s \in \mathbb{R}$ there is a constant C_s such that*

$$\|u\|_{H^{s+2}} \le C_s\big(\|\Delta_j u\|_{H^s} + \|u\|_{H^s}\big).$$

The closure of Δ_j initially defined on smooth sections is the self-adjoint operator with domain $H^2(E^j)$. It has compact resolvent. Hence its spectrum is a discrete sequence

$$0 \le \lambda_0 \le \lambda_1 \le \cdots \longrightarrow \infty$$

with finite multiplicities and a smooth orthonormal eigenbasis.

Proof. Choose a pseudodifferential parametrix Q of order -2 with $Q\Delta_j = I - R$, where R is smoothing. Mapping properties of Q and R give

$$\|u\|_{H^{s+2}} \le C\|\Delta_j u\|_{H^s} + C'\|u\|_{H^s}.$$

The symmetric nonnegative operator defined by the quadratic form

$$q_j(u) = \|D_j u\|^2 + \|D_{j-1}^* u\|^2, \qquad u \in H^1(E^j),$$

has a Friedrichs extension. The first-order elliptic estimate for $D_j \oplus D_{j-1}^*$ makes $q_j(u) + \|u\|^2$ equivalent to the H^1-norm; in particular the form is closed and its form domain really is $H^1(E^j)$. Elliptic regularity identifies its operator domain with $H^2(E^j)$, so this is the unique self-adjoint closure of the smooth Laplacian. For every $c > 0$, the energy identity makes $\Delta_j + c$ injective. Fredholmness and self-adjointness identify its cokernel with its kernel, hence it is bijective and its inverse maps L^2 boundedly to H^2. The inclusion $H^2 \hookrightarrow L^2$ is compact by Rellich's lemma, so $(\Delta_j + c)^{-1}$ is compact and self-adjoint on L^2. The spectral theorem for compact self-adjoint operators gives the asserted eigenbasis; elliptic regularity makes every eigenvector smooth. □

Theorem 2.11 (Hodge decomposition)**.** *For every j,*

$$L^2(E^j) = \mathcal{H}^j \,\widehat{\oplus}\, \operatorname{im} D_{j-1} \,\widehat{\oplus}\, \operatorname{im} D_j^*.$$

Here the operators are taken between their natural Sobolev domains; both ranges are closed. On smooth sections the decomposition restricts to

$$\Gamma(E^j) = \mathcal{H}^j \oplus D_{j-1}\Gamma(E^{j-1}) \oplus D_j^*\Gamma(E^{j+1}).$$

Moreover $\mathcal{H}^j$ is finite-dimensional and

$$H^j(D) \cong \mathcal{H}^j.$$

Proof. Let H_j be orthogonal projection onto the zero eigenspace. On the orthogonal complement define the Green operator by the convergent spectral series

$$G_j u = \sum_{\lambda_\nu > 0} \lambda_\nu^{-1} \langle u, e_\nu\rangle e_\nu, \qquad G_j H_j = H_j G_j = 0.$$

The first positive eigenvalue is separated from zero, so G_j is bounded on L^2, and

$$\Delta_j G_j = G_j \Delta_j = I - H_j.$$

The elliptic estimate applied repeatedly shows that G_j maps smooth sections to smooth sections.

On smooth sections the complex identities give

$$D_j\Delta_j = \Delta_{j+1}D_j, \qquad D_{j-1}^*\Delta_j = \Delta_{j-1}D_{j-1}^*.$$

Every eigenvector is smooth. Thus, if $\Delta_j e_\nu = \lambda_\nu e_\nu$, then $D_j e_\nu$ is either zero or a smooth λ_ν-eigenvector of Δ_{j+1}, and similarly for D_{j-1}^*. Applying the spectral definition of G on the finite linear span of the eigenvectors therefore gives

$$D_jG_j = G_{j+1}D_j, \qquad D_{j-1}^*G_j = G_{j-1}D_{j-1}^*.$$

These identities extend to their natural graph domains because that span is a core for all the operators involved. Moreover $G_j : L^2 \to H^2$ is bounded by the elliptic estimate.

Expanding $I - H_j = \Delta_j G_j$ now yields the explicit decomposition

$$u = H_j u + D_{j-1}D_{j-1}^*G_j u + D_j^*D_jG_j u.$$

All three displayed operators are bounded on L^2: the Green operator gains two derivatives and each remaining factor has total order two. The three terms are mutually orthogonal. For example,

$$\langle D_{j-1}v, D_j^*w\rangle = \langle D_jD_{j-1}v, w\rangle = 0.$$

Put

$$P_{\mathrm{ex}} = D_{j-1}D_{j-1}^*G_j, \qquad P_{\mathrm{coex}} = D_j^*D_jG_j.$$

The decomposition just proved is orthogonal, and $P_{\mathrm{ex}}u$ is exact while $P_{\mathrm{coex}}u$ is coexact. Applying the decomposition to either of these two terms shows

$$P_{\mathrm{ex}}^2 = P_{\mathrm{ex}}, \qquad P_{\mathrm{coex}}^2 = P_{\mathrm{coex}}, \qquad P_{\mathrm{ex}}P_{\mathrm{coex}} = 0.$$

Their ranges are exactly the Sobolev ranges in the theorem. Indeed, $P_{\mathrm{ex}}u = D_{j-1}(D_{j-1}^*G_ju)$ lies in $D_{j-1}H^1$; conversely, for $v = D_{j-1}w$, first with w smooth and then by H^1-approximation, orthogonality and $I = H_j + P_{\mathrm{ex}} + P_{\mathrm{coex}}$ give $P_{\mathrm{ex}}v = v$. The coexact case is identical. Ranges of bounded projections are closed. Compact resolvent makes $\mathcal{H}^j$ finite-dimensional, and elliptic regularity makes it smooth.

If $D_ju = 0$, pair u with its coexact summand:

$$\langle u, D_j^*w\rangle = \langle D_ju, w\rangle = 0.$$

Orthogonality therefore forces that summand to vanish, so u is cohomologous to H_ju. A harmonic exact form is zero, proving uniqueness. Finally, if u is smooth, then G_ju is smooth by elliptic regularity, so every term in the decomposition is smooth. This proves the smooth statement as well as the L^2 statement. □

2.3.1 The heat-kernel proof and the Green operator

The second approach starts from the heat operator. For $t > 0$, spectral calculus gives

$$e^{-t\Delta_j}u = \sum_\nu e^{-t\lambda_\nu}\langle u, e_\nu\rangle e_\nu.$$

It is smoothing, solves

$$(\partial_t + \Delta_j)e^{-t\Delta_j} = 0, \qquad \lim_{t\downarrow 0} e^{-t\Delta_j}u = u,$$

and converges in operator norm to H_j as $t \to \infty$. Therefore

$$G_j = \int_0^\infty (e^{-t\Delta_j} - H_j)\, dt.$$

The integral converges in operator norm because the nonzero spectrum is bounded below by the first positive eigenvalue. Integrating the heat equation gives

$$\Delta_j G_j = -\int_0^\infty \partial_t (e^{-t\Delta_j} - H_j)\, dt = I - H_j.$$

Thus the heat-parametrix construction in the next chapter does more than produce a fundamental solution: after integration in time it produces the same Green operator used in Hodge decomposition.

Question 2.12. *Does the heat-kernel proof assume Hodge decomposition when it uses the projection H_j?*

Answer. No. Compact resolvent is obtained from the elliptic estimate and Rellich compactness, independently of Hodge decomposition. The spectral theorem then defines the finite-dimensional zero-eigenspace and its orthogonal projection. Integrating the heat semigroup constructs G_j, and the identity $I - H_j = \Delta_j G_j$ yields Hodge decomposition. One may instead construct a heat parametrix first and use analytic Fredholm theory; the logical input is the same compactness statement. □

For the de Rham complex, with d^* given by (2.1) and $\Delta = dd^* + d^*d$, every de Rham cohomology class has a unique harmonic representative.

Corollary 2.13 (Poincaré duality)**.** *If M is closed and oriented of dimension n, then the pairing*

$$H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR}}(M) \longrightarrow \mathbb{R}, \qquad ([\alpha], [\beta]) \longmapsto \int_M \alpha \wedge \beta$$

is nondegenerate.

Proof. Choose the harmonic representative α of a nonzero class. The Hodge star commutes with the Laplacian, so $*\alpha$ is harmonic and represents a class in degree $n - k$. Then

$$\int_M \alpha \wedge *\alpha = \|\alpha\|^2_{L^2} > 0.$$

Hence no nonzero class lies in the left radical; symmetry gives the same conclusion for the right radical. □

Grade forms by parity and set

$$D_{\mathrm{dR}} = d + d^* : \Omega^{\mathrm{ev}}(M) \longrightarrow \Omega^{\mathrm{odd}}(M).$$

Hodge theory gives

$$\operatorname{ind} D_{\mathrm{dR}} = \chi(M).$$

2.4 The symbol class and the index theorem

Compactly supported K-theory, Thom and Bott classes

Definition 2.14. Let X be a locally compact space. An element of $K_c^0(X)$ is represented by a triple (E, F, σ) of complex vector bundles over X and a bundle map $\sigma : E \to F$ that is an isomorphism outside a compact subset of X. Two triples are identified if they are isomorphic, or homotopic through such triples with a common compact exceptional set; a triple in which σ is an isomorphism everywhere represents zero; and addition is direct sum. Equivalently, $K_c^0(X)$ is the relative group $K^0(\overline{X}, \overline{X} \setminus X)$ of any compactification. If X is compact, $K_c^0(X) = K^0(X)$ is the Grothendieck group of vector bundles.

An open inclusion $j : U \hookrightarrow X$ induces $j_! : K_c^0(U) \to K_c^0(X)$, extension by zero: a triple that is an isomorphism outside a compact subset of U is extended by the trivial triple, after stabilizing so that E and F extend.

Definition 2.15. Let $V \to X$ be a Hermitian vector bundle of complex rank r, with projection π. Its *Thom class* is

$$\lambda_V = \big[\pi^*\Lambda^{\mathrm{ev}}V,\ \pi^*\Lambda^{\mathrm{odd}}V,\ \varepsilon(v) - \iota_v\big] \in K_c^0(V),$$

where at $v \in V$ the map is exterior multiplication by v minus its adjoint. Since $(\varepsilon(v) - \iota_v)^*(\varepsilon(v) - \iota_v) = |v|^2$, it is an isomorphism off the zero section. For X a point and $V = \mathbb{C}^N$, the class $\beta_N = \lambda_{\mathbb{C}^N} \in K_c^0(\mathbb{C}^N)$ is the *Bott class*.

Theorem 2.16 (Thom isomorphism and Bott periodicity [1, 3])**.** *For X locally compact, $a \mapsto \pi^*a \cdot \lambda_V$ is an isomorphism $K_c^0(X) \to K_c^0(V)$. In particular $K_c^0(\mathbb{C}^N) \cong \mathbb{Z}$, generated by β_N.*

If $i : X \hookrightarrow Y$ is a closed embedding of manifolds whose normal bundle W carries a complex structure, identify W with an open tubular neighbourhood U of X. The *push-forward*

$$i_! : K_c^0(X) \longrightarrow K_c^0(Y), \qquad i_!(a) = j_!\big(\pi^*a \cdot \lambda_W\big),$$

is the Thom isomorphism followed by extension by zero along $j : U \hookrightarrow Y$. It is independent of the choices and functorial, $(i' \circ i)_! = i'_! \circ i_!$ [1].

The symbol class

If $P : E \to F$ is elliptic, its principal symbol is an isomorphism over $T^*M \setminus 0$. Hence the triple

$$(\pi^*E, \pi^*F, \sigma(P))$$

defines a compactly supported class

$$[\sigma(P)] \in K_c^0(T^*M).$$

Homotopic elliptic symbols determine the same class.

To construct the topological index, embed $i : M \hookrightarrow \mathbb{R}^N$ and use the metric to identify tangent and cotangent bundles. The differential of the embedding gives a closed embedding $T^*M \hookrightarrow T^*\mathbb{R}^N$. If ν is the normal bundle of M, the normal bundle of this induced embedding is $\pi^*(\nu \oplus \nu)$; the rule $J(v, w) = (-w, v)$ gives it a canonical complex structure. Write $i_! : K_c^0(T^*M) \to K_c^0(T^*\mathbb{R}^N)$ for the resulting push-forward, and identify $T^*\mathbb{R}^N = \mathbb{R}^N \times \mathbb{R}^N$ with $\mathbb{C}^N$ by $(x, \xi) \mapsto x + i\xi$. By Theorem 2.16, $i_!\alpha = k\,\beta_N$ for a unique integer k, and we set

$$\text{t-ind}(\alpha) = k.$$

Functoriality of push-forward and stabilization $\mathbb{R}^N \subset \mathbb{R}^{N+1}$ show that k is independent of the embedding. Note that $\beta_N = p_!(1)$ for the inclusion p of a point in $\mathbb{R}^N$, since the normal bundle of $T^*\{\mathrm{pt}\}$ in $T^*\mathbb{R}^N$ is $\mathbb{C}^N$.

Let $\mathcal{E}(M)$ denote stable homotopy classes of elliptic operators. The analytic and topological indices are compared in Figure 2.2.

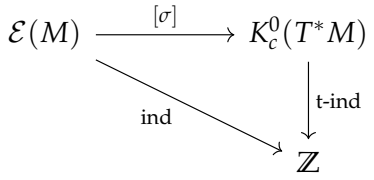


Figure 2.2: The Atiyah–Singer theorem states that the analytic index factors through the compactly supported symbol class.

Theorem 2.17 (Atiyah–Singer index theorem)**.** *Every elliptic operator P on a closed manifold M satisfies*

$$\operatorname{ind} P = \text{t-ind}[\sigma(P)].$$

In cohomological form,

$$\operatorname{ind} P = \epsilon_n \int_{T^*M} \operatorname{ch}[\sigma(P)] \wedge \pi^* \operatorname{Td}(T_{\mathbb{C}}M),$$

where $\operatorname{ch} : K_c^0(T^*M) \to H_c^{\mathrm{ev}}(T^*M;\mathbb{Q})$ *is the compactly supported Chern character and* $\epsilon_n = \pm 1$ *is a sign depending only on* $n = \dim M$ *and on the orientation convention for* T^*M *[2], [6, Chapter III].*

The first statement says that the analytic index depends only on the K-theory class of the principal symbol; it is proved in the next section, modulo the analytic push-forward theorem. The cohomological form is imported: it follows from the first by applying the Chern character, which converts the K-theoretic Thom isomorphism into the cohomological one at the cost of a Todd-class factor [2].

Discussion (From Dirac operators to general symbols). The heat-kernel chapters prove the cohomological formula, in the explicit form $\int_M \widehat{A}(TM)\operatorname{ch}(E/S)$, for Dirac operators of Clifford modules (Theorem 5.5). To pass to a general elliptic operator on a closed oriented even-dimensional manifold one uses two further imported facts [2], [6, Chapter III]: the classes $[\sigma(D_{\mathrm{sign}} \otimes W)]$ of twisted signature operators, $W \in K^0(M)$, span $K_c^0(T^*M) \otimes \mathbb{Q}$; and for these symbols the cohomological formula of Theorem 2.17 reduces to $\int_M \widehat{A}(TM)\operatorname{ch}(E/S)$. Since both indices are additive in the symbol class, the Dirac case then determines the general case. The odd-dimensional and nonorientable cases are reduced to this one by products with a fixed operator and by passing to the orientation double cover [2].

2.5 The K-theoretic proof

The preceding construction defines the topological index. We now explain why it equals the analytic index. The proof is short once two substantial facts have been isolated: an elliptic symbol can be quantized, and the analytic index commutes with push-forward along embeddings of closed manifolds. This is the K-theoretic proof of Atiyah and Singer [1, 3].

First define the analytic index directly on compactly supported K-theory, using the following lemma.

Lemma 2.18 (Elliptic representatives and quantization)**.** *After stabilization, every class $\alpha \in K_c^0(T^*M)$ has a representative*

$$\alpha = [\pi^*E, \pi^*F, \sigma]$$

in which $\sigma(x,\xi)$ is smooth on $T^*M \setminus 0$, homogeneous of degree zero, and invertible for every $\xi \neq 0$. It is the principal symbol of a classical operator $P_\sigma \in \Psi^0(M; E, F)$. If two such operators have the same principal symbol, then their difference is compact on L^2, so their Fredholm indices agree.

Proof. Let B^*M and S^*M be the closed unit ball and unit sphere bundles. Radial compactification gives

$$K_c^0(T^*M) \cong K^0(B^*M, S^*M).$$

A relative class is represented by a triple $(\widetilde{E}, \widetilde{F}, u)$, where $\widetilde{E}$ and $\widetilde{F}$ are bundles on B^*M and $u : \widetilde{E}|_{S^*M} \to \widetilde{F}|_{S^*M}$ is an isomorphism. Because B^*M deformation retracts onto its zero section, stable bundle isomorphisms identify $\widetilde{E}$ and $\widetilde{F}$ with π^*E and π^*F. Define on the punctured cotangent bundle

$$\sigma(x,\xi) = u\left(x, \frac{\xi}{|\xi|}\right).$$

It is smooth, homogeneous of degree zero, invertible at every nonzero covector, and represents the given relative class because its restriction to S^*M is u.

Choose $\chi \in C^\infty([0,\infty))$ equal to zero near the origin and one for $|\xi| \geq 1$. Then $\chi(|\xi|)\sigma(x,\xi)$, extended by zero near the zero section, is a smooth classical order-zero symbol with principal symbol σ. Classical quantization gives $P_\sigma \in \Psi^0(M; E, F)$, and the parametrix theorem makes it Fredholm. If P_σ and Q_σ have the same principal symbol, then

$$P_\sigma - Q_\sigma \in \Psi^{-1}(M; E, F).$$

This difference maps L^2 continuously to H^1, and the inclusion $H^1 \hookrightarrow L^2$ is compact by Rellich's lemma. Thus it is a compact perturbation and does not change the index. □

Set

$$\text{a-ind}(\alpha) = \operatorname{ind} P_\sigma.$$

Lemma 2.18 handles the choice of quantization. A homotopy of relative symbols can be quantized, using one fixed collection of coordinate charts and cutoffs, to a norm-continuous family of Fredholm operators on L^2; its index is constant. Direct sum gives additivity. For a stabilized summand whose boundary isomorphism extends over B^*M, radial contraction of the extension homotopes its boundary symbol to the pullback of an invertible bundle map on M. Quantizing that last map gives an invertible operator of index zero. These are exactly the relations defining $K^0(B^*M, S^*M)$, so a-ind is a well-defined homomorphism.

Theorem 2.19 (Analytic push-forward; imported)**.** *Let $i : M \hookrightarrow N$ be an embedding of closed manifolds, and let $i_! : K_c^0(T^*M) \to K_c^0(T^*N)$ be the push-forward along the induced embedding $T^*M \hookrightarrow T^*N$. For every $\alpha \in K_c^0(T^*M)$,*

$$\text{a-ind}(i_!\alpha) = \text{a-ind}(\alpha).$$

This is the analytic push-forward theorem of Atiyah–Singer [1]; it is an input to the proof below.

Discussion (What the analytic push-forward theorem supplies). Choose a tubular neighborhood with normal bundle ν. The normal bundle of T^*M in T^*N is $\pi^*(\nu \oplus \nu)$, with complex structure

$$J(v, w) = (-w, v).$$

Consequently $i_!$ is, in the tubular neighborhood, exterior product with the Bott–Thom class, followed by extension to T^*N.

Here is the analytic representative of that class. On a Euclidean fibre V act on $\Lambda^\bullet V_{\mathbb{C}}^*$ by

$$c(v) = \varepsilon(v^\flat) - \iota_v, \qquad \widehat{c}(v) = \varepsilon(v^\flat) + \iota_v.$$

For an orthonormal basis $e_1, \dots, e_r$, the odd Bott operator is

$$B_V = \sum_{j=1}^{r} c(e_j) \frac{\partial}{\partial v_j} + \widehat{c}(v).$$

Its full symbol in the position and covariable pair (v, ξ) is $i\,c(\xi) + \widehat{c}(v)$, the Bott class on $V \oplus V^*$. With the conventions above, c and $\widehat{c}$ anticommute, $\widehat{c}(v)^2 = |v|^2$, and

$$B_V^2 = \Delta_V + |v|^2 + \sum_{j=1}^{r} c(e_j)\widehat{c}(e_j), \qquad \Delta_V = -\sum_{j=1}^{r} \frac{\partial^2}{\partial v_j^2} \geq 0.$$

Here $c(e_j)\widehat{c}(e_j)$ equals $+1$ on forms containing $e_j^\flat$ and -1 otherwise, so the last sum is $2\mathcal{N} - r$, where $\mathcal{N}$ is the form degree, while $\Delta_V + |v|^2 \geq r$ with equality exactly on the Gaussian $e^{-|v|^2/2}$. Thus $B_V^2 \geq 2\mathcal{N}$, and the harmonic-oscillator decomposition shows that its kernel is the one-dimensional Gaussian vacuum in even Bott degree and that its odd kernel is zero. In particular, B_V is Fredholm and

$$\operatorname{ind} B_V = 1.$$

Write P_σ as the odd self-adjoint operator

$$\mathcal{P}_\sigma = \begin{pmatrix} 0 & P_\sigma^* \\ P_\sigma & 0 \end{pmatrix} \quad \text{on } E \oplus F.$$

Let $\Gamma_{E\oplus F}$ act by $+1$ on E and by -1 on F. In an exact product neighborhood with one fixed Euclidean normal fibre V, the graded product

$$\mathcal{P}_\sigma \widehat{\otimes} 1 + \Gamma_{E\oplus F} \widehat{\otimes} B_V$$

has symbol α multiplied by the Bott class. The two odd summands anticommute, so the square splits, and its kernel is the kernel of $\mathcal{P}_\sigma$ tensored with the Gaussian vacuum. Consequently the Fredholm product formula gives

$$\operatorname{ind}(P_\sigma \# B_V) = \operatorname{ind}(P_\sigma) \operatorname{ind}(B_V) = \operatorname{ind}(P_\sigma).$$

For a nontrivial normal bundle there is no single global fibre V and no fixed Gaussian vector with which the kernel can be tensorized. A metric and connection construct a family of vertical Bott operators B_v. Horizontal differentiation of that family and curvature of the connection produce mixed lower-order terms, so its square does not split. The family Bott theorem identifies its family index with the Thom unit; multiplicativity of the analytic index and tubular-neighborhood excision then give the displayed push-forward equality. A rescaling/Fredholm homotopy controls the mixed terms and proves independence of the connection and tubular neighborhood. These globalization, multiplicativity, and excision assertions are the content of the imported analytic push-forward theorem. See [1, 3] for the full family argument.

We can now prove the first statement of Theorem 2.17. The analytic index has been defined only on closed manifolds, so compactify: let $i : M \hookrightarrow \mathbb{R}^N$ be the embedding used to define t-ind, let $u : \mathbb{R}^N \hookrightarrow S^N$ be the inclusion as the complement of a point, and let $j : T^*\mathbb{R}^N \hookrightarrow T^*S^N$ be the induced open inclusion. Then $u \circ i$ is an embedding of closed manifolds, and by functoriality and excision of the topological push-forward, $(u \circ i)_! = j_! \circ i_!$. Let $p : \{\mathrm{pt}\} \to \mathbb{R}^N$ be the inclusion of a point; as noted above, $p_!(1) = \beta_N$, so $(u \circ p)_!(1) = j_!\beta_N$. Applying Theorem 2.19 to the closed embedding $u \circ p : \{\mathrm{pt}\} \hookrightarrow S^N$ gives

$$\text{a-ind}(j_!\beta_N) = \text{a-ind}(1) = 1,$$

because the operator $\mathbb{C} \to 0$ on a point has index one. If $i_!\alpha = k\beta_N$, then Theorem 2.19 applied to $u \circ i$, and additivity of a-ind, give

$$\text{a-ind}(\alpha) = \text{a-ind}\big(j_!i_!\alpha\big) = k\ \text{a-ind}(j_!\beta_N) = k = \text{t-ind}(\alpha).$$

Therefore a-ind $=$ t-ind on $K_c^0(T^*M)$, which proves the K-theoretic Atiyah–Singer theorem for every elliptic operator, given Theorem 2.19. The cohomological form and its specialization to Dirac symbols are the imported steps described after Theorem 2.17.

Question 2.20. *Where is the difficult geometry hidden in this short proof?*

Answer. It is concentrated in the imported push-forward theorem. Quantization converts symbol homotopy into Fredholm homotopy; the Thom class becomes the fibrewise Bott operator; and excision identifies a tubular neighborhood with its normal model. Once those compatibilities are supplied, the final reduction to $K_c^0(\mathbb{C}^N) \cong \mathbb{Z}$ is formal. □

Chapter 3

Heat kernels

3.1 Existence and basic properties

Definition 3.1. A second-order differential operator L on a vector bundle E over a Riemannian manifold is of *Laplace type* if its principal symbol is

$$\sigma_2(L)(x,\xi) = |\xi|^2 \operatorname{Id}_{E_x}.$$

Let L be a nonnegative, formally self-adjoint elliptic differential operator of order two on a Hermitian bundle $E \to M$. Its self-adjoint extension has discrete spectrum

$$0 \le \lambda_0 \le \lambda_1 \le \cdots, \qquad \lambda_j \longrightarrow \infty,$$

with a complete orthonormal basis of smooth eigensections $\{\phi_j\}$.

For $t > 0$, functional calculus defines

$$e^{-tL}u = \sum_j e^{-t\lambda_j} \langle u, \phi_j \rangle \phi_j.$$

Its Schwartz kernel is

$$K(t; x, y) = \sum_j e^{-t\lambda_j} \phi_j(x) \otimes \phi_j(y)^*.$$

The series converges in every smooth topology for $t \ge t_0 > 0$. Consequently e^{-tL} is smoothing and trace class, and

$$\operatorname{Tr}(e^{-tL}) = \sum_j e^{-t\lambda_j}.$$

The heat operator satisfies

$$(\partial_t + L)K = 0, \qquad \lim_{t \downarrow 0} e^{-tL}u = u,$$

and the semigroup identity

$$e^{-(t+s)L} = e^{-tL}e^{-sL}.$$

Proposition 3.2 (Duhamel formula)**.** *Let L_r be a differentiable family of nonnegative self-adjoint Laplace-type operators with common domain $H^2(E)$, and suppose $r \mapsto L_r$ is differentiable as a map $H^2(E) \to L^2(E)$. Then*

$$\frac{d}{dr} e^{-tL_r} = -\int_0^t e^{-(t-s)L_r} \dot{L}_r e^{-sL_r} \, ds.$$

Proof. Let $U_r(t)$ denote the right side. Differentiation of $(\partial_t + L_r)e^{-tL_r} = 0$ with respect to r shows that $\partial_r e^{-tL_r}$ solves

$$(\partial_t + L_r)U = -\dot{L}_r e^{-tL_r}, \qquad U(0) = 0.$$

Variation of constants gives the displayed integral. Uniqueness of the inhomogeneous heat equation completes the proof. □

3.2 The local parametrix

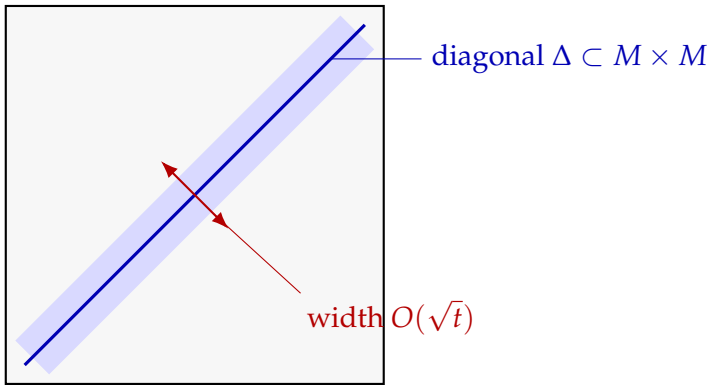
From now on L is of Laplace type (Definition 3.1). Every Laplace-type operator has a unique expression

$$L = \nabla^*\nabla + \mathcal{R}$$

for a connection ∇ and an endomorphism $\mathcal{R}$.

Near the diagonal, let $r(x,y)$ be the Riemannian distance. Seek a parametrix of the form

$$H_N(t;x,y) = (4\pi t)^{-n/2} e^{-r(x,y)^2/(4t)} \sum_{j=0}^{N} t^j u_j(x,y).$$



$e^{-d(x,y)^2/(4t)}$ localizes near the diagonal

Figure 3.1: As $t \downarrow 0$, the heat kernel concentrates in a $\sqrt{t}$-neighborhood of the diagonal; the transport equations resolve its coefficients along radial geodesics.

Substitution into $(\partial_t + L_x)H_N = 0$ first determines u_0 by parallel transport and the Riemannian density, then determines each u_j from a transport equation along the radial geodesic.

More explicitly, let $\gamma(\rho) = \exp_y(\rho v)$, $0 \le \rho \le 1$, with $x = \exp_y v$. If $J(x,y)$ is the Jacobian of the exponential map and $\tau(x,y) : E_y \to E_x$ is parallel transport, then

$$u_0(x,y) = J(x,y)^{-1/2}\tau(x,y), \qquad u_0(y,y) = I.$$

After conjugation by u_0, the j-th transport equation is an ordinary radial equation. Its solution is

$$u_j(x,y) = -u_0(x,y) \int_0^1 \rho^{j-1} u_0(\gamma(\rho),y)^{-1} (L_x u_{j-1})(\gamma(\rho),y)\, d\rho, \qquad j \ge 1.$$

The integral is smooth at $x = y$, and induction gives smooth u_j on a normal neighborhood of the diagonal.

Multiply by a cutoff supported in a normal neighborhood of the diagonal. For every pair of nonnegative integers a, b, the error

$$R_N = (\partial_t + L)H_N$$

satisfies, after solving sufficiently many transport equations,

$$\|\partial_t^a R_N(t)\|_{C^b(M\times M)} \le C_{a,b,N} t^{N-n/2-a-b/2}.$$

Each spatial derivative can cost a factor $t^{-1/2}$; the stronger estimate without $b/2$ would be false for a fixed truncation. For N large relative to the requested seminorms, the Volterra series

$$K = H_N - H_N * R_N + H_N * R_N * R_N - \cdots$$

converges in the Fréchet topology of smooth kernels on $(0, T] \times M \times M$, with its natural weighted seminorms at $t = 0$, where

$$(A * B)(t) = \int_0^t A(t-s)B(s)\, ds.$$

It solves the heat equation with the correct initial value and therefore equals the heat kernel.

The algebraic cancellation uses the distributional identity $H_N(0) = I$. For any residual kernel S, differentiation of the upper convolution limit gives

$$(\partial_t + L)(H_N * S) = S + R_N * S.$$

Consequently successive terms in the Volterra series cancel pairwise. This boundary term is essential; without it the series would be only a formal Neumann expression.

For completeness, the Volterra series converges because its k-fold term is integrated over an ordered simplex. If $\|R_N(t)\| \le Ct^\alpha$, $\alpha > -1$, then repeated convolution gives

$$\|R_N^{*k}(t)\| \le \frac{C^k \Gamma(\alpha+1)^k}{\Gamma(k(\alpha+1))} t^{k(\alpha+1)-1}.$$

The gamma function supplies factorial decay. The same estimate for derivatives gives convergence in every smooth kernel seminorm.

Theorem 3.3 (Heat-kernel expansion)**.** *On the diagonal,*

$$K(t; x, x) \sim (4\pi t)^{-n/2} \sum_{j=0}^{\infty} t^j a_j(x) \qquad (t \downarrow 0),$$

in the smooth topology. Each $a_j(x)$ is a universal polynomial in the curvatures of M and E, the endomorphism $\mathcal{R}$, and their covariant derivatives. In particular,

$$a_0 = \mathrm{Id}_E, \qquad a_1 = \frac{1}{6}\,\mathrm{scal}\ \mathrm{Id}_E - \mathcal{R}$$

for the convention $L = \nabla^*\nabla + \mathcal{R}$.

Integrating the fibrewise trace gives

$$\mathrm{Tr}(e^{-tL}) \sim (4\pi t)^{-n/2} \sum_{j=0}^{\infty} t^j \int_M \mathrm{tr}_E\, a_j(x)\ d\,\mathrm{vol}(x).$$

3.3 McKean–Singer

Let

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}$$

be an odd, formally self-adjoint elliptic operator on a graded bundle $E^+ \oplus E^-$. Then

$$D^2 = \begin{pmatrix} D^-D^+ & 0 \\ 0 & D^+D^- \end{pmatrix}.$$

For a trace-class endomorphism T, define

$$\operatorname{Str}(T) = \operatorname{Tr}(T|_{E^+}) - \operatorname{Tr}(T|_{E^-}).$$

Theorem 3.4 (McKean–Singer)**.** *For every $t > 0$,*

$$\operatorname{Str}(e^{-tD^2}) = \operatorname{ind} D^+.$$

Proof. Every positive eigenvalue λ of D^-D^+ is paired with the same eigenvalue of D^+D^-: the map D^+ is an isomorphism between the corresponding eigenspaces, with inverse $\lambda^{-1}D^-$. Their heat traces cancel in the supertrace. The zero eigenspaces contribute $\dim\ker D^+ - \dim\ker D^-$, which is the index. □

Question 3.5. *Why do the zero modes survive while every positive mode cancels?*

Answer. On a positive eigenspace, D^+ has inverse $\lambda^{-1}D^-$, so the even and odd multiplicities are equal and carry the same heat weight $e^{-t\lambda}$. At eigenvalue zero this inverse does not exist. The unpaired spaces are precisely $\ker D^+$ and $\ker D^-$, whose signed dimension is the index. □

Equivalently, cyclicity of the supertrace gives

$$\frac{d}{dt}\operatorname{Str}(e^{-tD^2}) = -\frac{1}{2}\operatorname{Str}([D, De^{-tD^2}]_{\mathrm{s}}) = 0.$$

The small-time limit is therefore the index.

When the pointwise limit exists and

$$\operatorname{str} K(t;x,x) \sim (4\pi t)^{-n/2} \sum_{j\geq 0} t^j \operatorname{str} a_j(x),$$

then

$$\operatorname{ind} D^+ = \int_M \lim_{t\downarrow 0} \operatorname{str} K(t;x,x)\ d\operatorname{vol}(x).$$

For an arbitrary elliptic operator $P : \Gamma(E) \to \Gamma(F)$ of order k, McKean–Singer still applies to the odd self-adjoint operator

$$\mathcal{D} = \begin{pmatrix} 0 & P^* \\ P & 0 \end{pmatrix}, \qquad \operatorname{ind} P = \operatorname{Str}\big(e^{-t\mathcal{D}^2}\big) \quad (t > 0),$$

because $\mathcal{D}^2 = P^*P \oplus PP^*$ is a nonnegative self-adjoint elliptic operator with compact resolvent. The local formula above, however, uses the expansion of Theorem 3.3, which was constructed only for Laplace-type operators. If $k \neq 1$, or if $\mathcal{D}^2$ is not of Laplace type, the diagonal heat kernel has an expansion in the powers $t^{(j-n)/2k}$ of a different parametrix, and we do not use it. The remaining task is to identify the constant term of the local supertrace. For Dirac operators, Clifford cancellation together with Getzler's filtration bound forces the singular terms to cancel, and Getzler rescaling computes the survivor.

Chapter 4

Clifford modules and Dirac operators

4.1 Clifford algebras

For a Euclidean vector space V, the complex Clifford algebra is

$$\mathrm{Cl}(V) = T(V_{\mathbb{C}}) \big/ \langle vw + wv + 2\langle v, w\rangle \rangle.$$

Thus

$$c(v)c(w) + c(w)c(v) = -2\langle v, w\rangle.$$

The algebra is $\mathbb{Z}_2$-graded by tensor parity.

If $\dim V = 2m$, the complex spin representation S has dimension 2^m and a grading

$$S = S^+ \oplus S^-.$$

Clifford multiplication by a vector is odd. The grading operator is the complex volume element, normalized to square to one and to agree with the chosen orientation.

Definition 4.1. A graded Clifford module over an oriented even-dimensional Riemannian manifold is a graded Hermitian bundle

$$E = E^+ \oplus E^-$$

with a fibrewise action

$$c : \mathrm{Cl}(T^*M) \longrightarrow \mathrm{End}(E)$$

such that $c(\xi)$ is odd and skew-adjoint for real ξ.

Locally, every complex Clifford module in even dimension is of the form

$$E \cong S \otimes W,$$

where the Clifford algebra acts on S and trivially on the twisting factor W. The bundle W need not exist globally unless a spinor bundle does, but its Chern character is globally represented by the relative curvature introduced below.

4.2 Clifford connections

Definition 4.2. A connection ∇^E on a graded Clifford module E is a *Clifford connection* if it is unitary for the Hermitian metric, preserves the grading $E = E^+ \oplus E^-$, and satisfies

$$[\nabla^E_X, c(\alpha)] = c(\nabla^{T^*M}_X \alpha)$$

for every vector field X and one-form α, where ∇^{T^*M} is the Levi–Civita connection.

The associated Dirac operator is

$$D = c \circ \nabla^E = \sum_{j=1}^{n} c(e^j) \nabla^E_{e_j}$$

in a local orthonormal frame. It is odd because $c(e^j)$ is odd and ∇^E preserves the grading. It is formally self-adjoint: for compactly supported u, v, define the one-form $\theta(X) = \langle c(X^\flat)u, v\rangle$. At the centre of a synchronous frame, unitarity, skew-adjointness of $c(e^j)$, and the Clifford condition give

$$\langle Du, v\rangle - \langle u, Dv\rangle = \sum_j e_j \langle c(e^j)u, v\rangle = -d^*\theta,$$

whose integral vanishes. Its principal symbol is

$$\sigma_1(D)(x, \xi) = i\, c(\xi).$$

Since

$$(i\, c(\xi))^2 = |\xi|^2 \,\mathrm{Id},$$

the operator is elliptic.

The curvature of a Clifford connection splits into its Clifford and twisting parts. Define

$$F^{E/S}(X, Y) = F^E(X, Y) - \frac{1}{4} \sum_{j,k} \langle R^{TM}(X, Y)e_j, e_k\rangle c(e^j)c(e^k).$$

This endomorphism commutes with Clifford multiplication. Locally, for $E = S \otimes W$, it is exactly F^W.

For an endomorphism-valued two-form F, write

$$c(F) = \frac{1}{2} \sum_{j,k} c(e^j)c(e^k)F(e_j, e_k).$$

Theorem 4.3 (Weitzenböck–Lichnerowicz formula)**.** *A Dirac operator satisfies*

$$D^2 = (\nabla^E)^*\nabla^E + \frac{1}{4}\,\mathrm{scal} + c(F^{E/S}).$$

Proof. At a chosen point, take a synchronous orthonormal frame, so $\nabla e_j = 0$ there. The symmetric part of $\sum_{j,k} c(e^j)c(e^k)\nabla_{e_j}\nabla_{e_k}$ is the connection Laplacian. The antisymmetric part is

$$\frac{1}{2} \sum_{j,k} c(e^j)c(e^k)F^E(e_j, e_k).$$

Insert the decomposition of F^E. The first Bianchi identity and the Clifford relations contract the Riemannian curvature term to scal $/4$; the remainder is $c(F^{E/S})$.

For the contraction, write the spin-curvature contribution as

$$\frac{1}{8}\sum_{i,j,k,\ell} R_{ijk\ell}\, c(e^i)c(e^j)c(e^k)c(e^\ell).$$

Terms in which i, j, k are distinct cancel by the first Bianchi identity. In the remaining terms $k \in \{i, j\}$; using the skew symmetries of R and $c(e^i)^2 = -1$, they add up to $-\frac{1}{4}\sum_{i,\ell}\big(\sum_j R_{ijj\ell}\big)c(e^i)c(e^\ell)$. The inner sum is symmetric in i, ℓ by pair symmetry, so only $i = \ell$ survives, and the total is

$$\frac{1}{4}\sum_{i,j} R_{ijji} = \frac{1}{4}\,\mathrm{scal}\,.$$

□

4.3 Basic examples

On differential forms, define

$$c(\xi) = \varepsilon(\xi) - \iota_{\xi^\sharp}.$$

The exterior and contraction operators satisfy

$$\varepsilon(\xi)\iota_{\eta^\sharp} + \iota_{\eta^\sharp}\varepsilon(\xi) = \langle \xi, \eta\rangle,$$

so c obeys the Clifford relation. The Levi–Civita connection on forms is a Clifford connection, and its Dirac operator is

$$D_{\mathrm{dR}} = d + d^*.$$

Its grading by even and odd degree gives

$$\operatorname{ind} D^+_{\mathrm{dR}} = \chi(M).$$

Example 4.4 (The Cauchy–Riemann operator as a Dirac operator)**.** On $\mathbb{R}^2 \cong \mathbb{C}$, with $z = x + iy$, set

$$\bar\partial = \frac{1}{2}(\partial_x + i\partial_y), \qquad \bar\partial^* = -\frac{1}{2}(\partial_x - i\partial_y).$$

Using the orthonormal frame $d\bar z/\sqrt{2}$ for $(0,1)$-forms, the Dolbeault–Dirac operator has the scalar matrix

$$\mathcal{D}_{\bar\partial} = 2\begin{pmatrix} 0 & \bar\partial^* \\ \bar\partial & 0 \end{pmatrix}$$

and principal symbol

$$\sigma_1(\mathcal{D}_{\bar\partial})(\xi) = i\begin{pmatrix} 0 & -(\xi_x - i\xi_y) \\ \xi_x + i\xi_y & 0 \end{pmatrix},$$

whose square is $|\xi|^2$ times the identity. This is the two-dimensional spin^c Dirac symbol and explains why the Dolbeault operator later has the same elliptic symbol as a Dirac operator.

If M is spin, its spinor bundle $S = S^+ \oplus S^-$ carries the spin connection and spin Dirac operator

$$D_S : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

For a Hermitian bundle W with unitary connection, the twisted operator on $S \otimes W$ satisfies

$$D_W^2 = \nabla^*\nabla + \frac{1}{4}\operatorname{scal} + c(F^W).$$

Corollary 4.5 (Lichnerowicz vanishing)**.** *If M is closed, spin, and has strictly positive scalar curvature, then the untwisted spin Dirac operator has zero kernel and zero index.*

Proof. For $D_S\psi = 0$, integration of the Lichnerowicz formula gives

$$0 = \|\nabla\psi\|^2 + \frac{1}{4}\int_M \operatorname{scal}|\psi|^2.$$

Strict positivity forces $\psi = 0$. □

4.4 Clifford supertrace

The supertrace on a graded endomorphism algebra is

$$\operatorname{str}(T) = \operatorname{tr}(T|_{E^+}) - \operatorname{tr}(T|_{E^-}).$$

It vanishes on supercommutators.

Lemma 4.6 (Clifford cancellation)**.** *Let $\dim V = n$ be even and let S be the spin representation. The supertrace of an element of reduced Clifford degree less than n is zero. Equivalently, the supertrace vanishes below degree n in the associated graded Clifford algebra. On the top Clifford degree, it is the oriented volume functional, with normalization fixed by the chirality operator.*

Proof. Use the Clifford relations first to reduce every word to a linear combination of products of distinct generators. Odd reduced monomials are odd endomorphisms and therefore have zero supertrace. Let $T = c(e^{i_1}) \cdots c(e^{i_{2r}})$ have distinct indices and even degree $2r < n$, and choose e^j not among its factors. Then $c(e^j)$ commutes with T but anticommutes with the chirality operator Γ. Since $\operatorname{str}(T) = \operatorname{tr}(\Gamma T)$, ordinary trace invariance under conjugation by $c(e^j)$ gives

$$\operatorname{str}(T) = \operatorname{tr}\Big(c(e^j)^{-1}\Gamma T c(e^j)\Big) = -\operatorname{str}(T),$$

so it vanishes. In degree n, the monomial is a scalar multiple of the complex volume element and hence of Γ; its supertrace is nonzero. The chosen orientation and the relation $c(e^j)^2 = -1$ fix that scalar. □

Discussion. Clifford cancellation alone does not remove the negative powers of t in the local heat supertrace: the heat coefficients $a_j(x)$ of Theorem 3.3 for D^2 are, a priori, arbitrary elements of $\operatorname{Cl}(T_x^*M) \otimes \operatorname{End}_{\operatorname{Cl}}(E_x)$. The additional input is *Getzler's filtration bound*: $a_j(x)$ has Clifford filtration degree at most $2j$ [4, Chapter 4]. Combined with Lemma 4.6, this gives $\operatorname{str} a_j(x) = 0$ for $2j < n$, pointwise and before integration, so $\operatorname{str} K(t; x, x) = \operatorname{str} a_{n/2}(x)(4\pi)^{-n/2} + O(t)$. McKean–Singer says the integral is independent of t; these two facts explain why its small-time limit is already a smooth top-degree differential form.

Chapter 5

The local index theorem

5.1 Relative curvature

Let $\dim M = 2m$, let Γ_E be the given module grading, and let Γ_{Cl} be the chirality operator obtained from the oriented complex Clifford volume element. Both anticommute with multiplication by one-forms, so $\Gamma_E\Gamma_{\mathrm{Cl}}$ lies in the Clifford commutant.

Lemma 5.1 (Intrinsic relative trace)**.** *For every even $A \in \mathrm{End}_{\mathrm{Cl}}(E)$, set*

$$\mathrm{tr}_{E/S}(A) = 2^{-m}\,\mathrm{tr}_E\big(\Gamma_E\Gamma_{\mathrm{Cl}}A\big).$$

This definition is intrinsic. Under a local graded Morita factorization $E \cong S\widehat{\otimes}W$, it becomes

$$\mathrm{tr}_{E/S}(I_S \otimes T) = \mathrm{str}_W(T).$$

Consequently it is independent of the choice of local spinor factorization.

Proof. In the displayed factorization,

$$\Gamma_E = \Gamma_S \otimes \Gamma_W, \qquad \Gamma_{\mathrm{Cl}} = \Gamma_S \otimes I_W.$$

Hence $\Gamma_E\Gamma_{\mathrm{Cl}} = I_S \otimes \Gamma_W$, and

$$2^{-m}\,\mathrm{tr}_{S\otimes W}\big(I_S \otimes \Gamma_W T\big) = \mathrm{tr}_W(\Gamma_W T) = \mathrm{str}_W(T),$$

because $\dim_{\mathbb{C}} S = 2^m$. The expression on the left uses only the global Clifford action, orientation, and module grading. Thus all local Morita factorizations give the same functional. □

For a graded Clifford module E, define the relative Chern character form by

$$\mathrm{ch}(E/S, \nabla^E) = \mathrm{tr}_{E/S}\exp\left(\frac{iF^{E/S}}{2\pi}\right).$$

Locally this is the graded Chern character of W; when W is ungraded, it is the ordinary $\mathrm{ch}(W)$. Since $F^{E/S}$ lies in the Clifford commutant and satisfies its own Bianchi identity, Chern–Weil transgression proves that this form is closed and that its cohomology class is independent of the Clifford connection.

The local index density is

$$\Big[\widehat{A}(TM, \nabla^{TM})\,\mathrm{ch}(E/S, \nabla^E)\Big]_{[n]}.$$

5.2 Getzler rescaling

Fix $p \in M$. Choose geodesic normal coordinates centered at p and trivialize E by radial parallel transport. In these coordinates, the connection and metric have expansions whose first nonconstant terms are determined by curvature.

In radial gauge, with the convention (1.1),

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3} R_{ik\ell j}(p) x^k x^\ell + O(|x|^3), \qquad \nabla_i^E = \partial_i - \frac{1}{2} F_{ij}^E(p) x^j + O(|x|^2).$$

For the unit sphere the first formula reads $g_{ij} = \delta_{ij} - \frac{1}{3}(|x|^2 \delta_{ij} - x^i x^j) + O(|x|^3)$, in agreement with $g = dr^2 + \sin^2 r \, d\theta^2_{S^{n-1}}$. The second formula follows by differentiating the radial-gauge condition $x^i \nabla_i^E = x^i \partial_i$: the linear part of the connection one-form is forced by its curvature. Splitting F^E into spin and relative curvature separates the part that acts through Clifford multiplication from the part commuting with it.

Assign Getzler degrees

$$\deg \partial_j = 1, \qquad \deg c(dx^j) = 1, \qquad \deg x^j = -1, \qquad \deg \partial_t = 2.$$

Under the parabolic scaling

$$x \mapsto \sqrt{t}\, x,$$

only the highest Getzler-order part of $\partial_t + D^2$ contributes to the constant term of the Clifford supertrace. The Clifford filtration is identified at its associated graded level with the exterior algebra:

$$c(dx^{i_1}) \cdots c(dx^{i_k}) \longmapsto dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

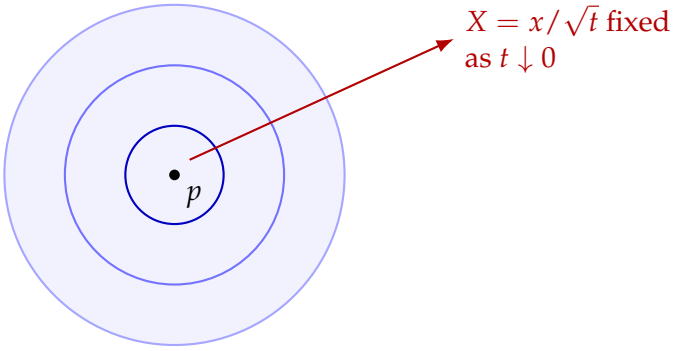


Figure 5.1: Getzler rescaling magnifies a shrinking neighborhood of p. The metric becomes Euclidean while the first nontrivial connection terms retain the curvature.

The rescaled model of D^2 is a harmonic oscillator with coefficients given by R_p^{TM}, together with the constant twisting curvature $F_p^{E/S}$. All terms of lower Getzler order disappear in the limit.

To state the model, put

$$\mathcal{R}_{ij} = \frac{1}{2} \sum_{k,\ell} R_{ijk\ell}(p)\, dx^k \wedge dx^\ell.$$

After passing from the Clifford filtration to its exterior associated graded, the model operator is

$$\mathcal{H}_p = -\sum_i \left(\partial_i - \frac{1}{4}\sum_j \mathcal{R}_{ij}x^j\right)^2 + F_p^{E/S}.$$

Here the curvature two-forms act by exterior multiplication in the associated graded Clifford algebra, while $F_p^{E/S}$ acts on the Clifford commutant. It is useful to separate the raw curvature in this differential operator from the normalized Chern–Weil variables

$$\Omega^{TM} = \frac{R^{TM}}{2\pi i}, \qquad \Omega^{E/S} = \frac{F^{E/S}}{2\pi i}. \tag{5.1}$$

With the convention of this manuscript,

$$\widehat{A}(TM) = \det{}^{1/2}\left(\frac{\Omega^{TM}/2}{\sinh(\Omega^{TM}/2)}\right), \qquad \mathrm{ch}(E/S) = \mathrm{tr}_{E/S}\, e^{-\Omega^{E/S}},$$

because $iF/(2\pi) = -F/(2\pi i)$. The scalar-curvature term in the Lichnerowicz formula has lower Getzler degree and disappears; its effect is already encoded in the oscillator through R_p^{TM}. Define $\delta_\varepsilon u(x) = u(\varepsilon x)$. Direct calculation gives

$$\delta_\varepsilon \partial_i \delta_\varepsilon^{-1} = \varepsilon^{-1}\partial_i, \qquad \delta_\varepsilon x^i \delta_\varepsilon^{-1} = \varepsilon x^i.$$

Let $\mathcal{C}_\varepsilon$ denote the Rees rescaling of the Clifford filtration: after applying the Clifford symbol map, its exterior-degree k component is multiplied by ε^{-k}. This is a rescaling on the Rees family, not an automorphism of one fixed Clifford algebra. With both rescalings displayed, the rescaled operator satisfies

$$\mathcal{C}_\varepsilon\left(\varepsilon^2 \delta_\varepsilon D^2 \delta_\varepsilon^{-1}\right) = \mathcal{H}_p + O(\varepsilon) \tag{5.2}$$

in the associated graded, coefficientwise on compact sets. Indeed, a linear spin-connection term has the form $xR_{ij}c(e^i)c(e^j)$: coordinate dilation contributes ε, Clifford degree two contributes ε^{-2}, and it therefore has the same ε^{-1} scale as ∂_i before the outer factor ε^2 is applied to the square. The term $c(F^{E/S})$ has Clifford degree two and likewise survives as $F_p^{E/S}$. Every term of lower Getzler order gains a positive power of ε. The order of conjugation matters: since $\delta_\varepsilon \partial_i \delta_\varepsilon^{-1} = \varepsilon^{-1}\partial_i$, the factor ε^2 exactly compensates the second-order part.

Lemma 5.2 (Normalized model heat kernel; Mehler formula imported)**.** *Let $n = 2m$, and use the curvature and chirality conventions fixed above. The scalar Mehler formula for the model operator gives, in the exterior associated graded,*

$$K_{\mathcal{H}_p}(t;0,0) = (4\pi t)^{-m} \det{}^{1/2}\left(\frac{tR_p^{TM}/2}{\sinh(tR_p^{TM}/2)}\right) e^{-tF_p^{E/S}}. \tag{5.3}$$

The square root is the formal branch with constant term one. After the top Clifford supertrace, its degree-n diagonal value is

$$\left[\widehat{A}(TM,\nabla^{TM})\,\mathrm{ch}(E/S,\nabla^E)\right]_{[n]}(p).$$

Normalization calculation. Formally put the skew-symmetric curvature matrix into oriented 2×2 blocks. If r denotes the formal curvature root of one block, its oscillator heat kernel at the origin contributes

$$(4\pi t)^{-1}\frac{tr/2}{\sinh(tr/2)}.$$

Multiplying the m block formulas gives the determinant and the factor $(4\pi t)^{-m}$ in (5.3). Because $F_p^{E/S}$ commutes with the Clifford action, it commutes with every oscillator block and contributes $e^{-tF_p^{E/S}}$. This blockwise Mehler identity, including its Fourier normalization, is the analytic formula imported from [4, Chapter 4].

It remains to check the Clifford and Chern–Weil constants. In an oriented orthonormal basis, set

$$C = c(e^1)\cdots c(e^{2m}), \qquad \Gamma_{\mathrm{Cl}} = i^m C.$$

Since $C^2 = (-1)^m$ and $\dim_{\mathbb{C}} S = 2^m$,

$$\mathrm{str}_S(C) = \mathrm{tr}_S(\Gamma_{\mathrm{Cl}} C) = 2^m(-i)^m. \tag{5.4}$$

Lower Clifford degrees have zero supertrace. Every exterior-degree $2m$ term on the right of (5.3) contains exactly m curvature two-forms and hence carries t^m. The powers of t cancel, and the remaining scalar factor is

$$(4\pi)^{-m}2^m(-i)^m = \left(\frac{1}{2\pi i}\right)^m.$$

Thus the top supertrace replaces each raw curvature two-form R by its normalized value $R/(2\pi i)$. With the normalization (5.1), the raw factor $e^{-tF^{E/S}}$ becomes $e^{-\Omega^{E/S}} = e^{iF^{E/S}/2\pi}$, and the oscillator determinant becomes

$$\det{}^{1/2}\left(\frac{\Omega^{TM}/2}{\sinh(\Omega^{TM}/2)}\right).$$

Taking the relative trace on the commutant and extracting total exterior degree n therefore gives

$$\left[\widehat{A}(TM,\nabla^{TM})\,\mathrm{ch}(E/S,\nabla^E)\right]_{[n]}(p).$$

This accounts explicitly for the top Clifford-supertrace constant, the powers of t and 4π, the factors of i and 2π, and the extraction of exterior degree n. The only imported part of this lemma is the blockwise scalar Mehler formula (5.3) itself. □

The determinant factor is $\widehat{A}(TM)$, and the relative trace of the constant curvature exponential is $\mathrm{ch}(E/S)$.

5.3 The pointwise theorem

Theorem 5.3 (Local index theorem for Dirac operators)**.** *Let D be the Dirac operator of a graded Clifford module E over a closed even-dimensional Riemannian manifold. If $K(t;x,y)$ is the heat kernel of D^2, then*

$$\lim_{t\downarrow 0}\mathrm{str}_E\,K(t;x,x)\;d\,\mathrm{vol}(x) = \left[\widehat{A}(TM,\nabla^{TM})\,\mathrm{ch}(E/S,\nabla^E)\right]_{[n]}(x).$$

Proof from the rescaled heat-kernel theorem. We use the standard rescaled heat-kernel theorem: in synchronous gauge, the Getzler-rescaled kernel of D^2 converges with all derivatives on compact sets to the kernel of $\mathcal{H}_p$, and the rescaled remainder has uniform Gaussian bounds. This is the analytic core of the local index theorem; complete constructions are given in [4, Chapters 4–5] and [5]. The ordinary heat parametrix by itself does not imply this uniform rescaled statement.

Given that theorem, the Clifford-cancellation lemma shows that coefficients of Getzler degree below n have zero supertrace. After parabolic rescaling, the only surviving coefficient is the heat

kernel of the curvature model. Lemma 5.2 and the Clifford top-degree supertrace identify it with the displayed Chern–Weil form.

For the remainder, choose enough terms in the heat parametrix that its error is $O(t^N)$ in every required C^k-seminorm. Under $x = \sqrt{t}\,X$, a term of Getzler order $n - j$ gains $t^{j/2}$. Terms killed by supertrace vanish identically, and every surviving term other than the model therefore tends to zero. The Volterra correction has the same estimate by the ordered-simplex bound from the heat-kernel construction. This proves smooth convergence of the local density, not merely convergence after integration. □

Question 5.4. *Why does the supertrace retain only the top exterior degree?*

Answer. Getzler rescaling turns the Clifford filtration into the exterior algebra. The fibrewise supertrace vanishes on every Clifford degree below n, by Lemma 4.6. Degree above n is zero in $\Lambda^\bullet T_p^* M$. Thus only degree n survives, and that degree is naturally an n-form that can be integrated over M. □

Integrating Theorem 5.3 and applying McKean–Singer (Theorem 3.4) gives the global formula.

Theorem 5.5 (Dirac index theorem)**.** *Under the same hypotheses,*

$$\boxed{\operatorname{ind} D^+ = \int_M \left[\widehat{A}(TM)\operatorname{ch}(E/S)\right]_{[n]}.}$$

For a spin manifold and $E = S \otimes W$,

$$\operatorname{ind} D_W^+ = \int_M \left[\widehat{A}(TM)\operatorname{ch}(W)\right]_{[n]}.$$

For W trivial, this is the $\widehat{A}$-genus.

5.4 Metric and connection independence

The left side of the index formula is constant under continuous deformations through elliptic operators. On the right, the transgression formula shows that a change of connection alters each characteristic form by an exact form. Since M is closed, its integral is unchanged.

This agreement separates the two roles in the proof: heat-kernel analysis identifies a local curvature density, while Chern–Weil theory proves that its integral is topological.

Chapter 6

Gauss–Bonnet and the signature theorem

6.1 The Euler operator

On an oriented Riemannian manifold, the de Rham operator

$$D_{\mathrm{dR}} = d + d^*$$

is the Dirac operator on the Clifford module $\Lambda^\bullet T^*M_{\mathbb{C}}$, graded by parity.

For the two characteristic-form computations in this chapter we need the relative Chern character of $\Lambda^\bullet T^*M_{\mathbb{C}}$ for two different gradings. Let $x_1, \dots, x_m$ be the Chern roots of TM, with the convention fixed in Chapter 1, so that $R^{TM}/2\pi$ has the formal blocks $\left(\begin{smallmatrix} 0 & x_j \\ -x_j & 0 \end{smallmatrix}\right)$.

Lemma 6.1 (Two-plane computation)**.** *Let Γ_{par} be the parity grading of $\Lambda^\bullet T^*M_{\mathbb{C}}$, and let*

$$\Gamma_{\mathrm{Cl}} = i^m c(e^1) \cdots c(e^{2m})$$

be the Clifford chirality. Then

$$\mathrm{ch}_{\mathrm{par}}(\Lambda^\bullet/S) = \prod_{j=1}^{m} 2\sinh(x_j/2), \qquad \mathrm{ch}_{\Gamma_{\mathrm{Cl}}}(\Lambda^\bullet/S) = \prod_{j=1}^{m} 2\cosh(x_j/2),$$

where the subscript indicates the module grading used in the relative trace.

Proof. For this module $F^{E/S}$ is a linear function of R^{TM}, so both sides are invariant polynomials in R^{TM} and may be evaluated on a formally block-diagonal curvature, i.e. as if $T_xM = L_1 \oplus \cdots \oplus L_m$ orthogonally with the curvature preserving each plane. Then $\Lambda^\bullet T_x^*M_{\mathbb{C}}$ is the graded tensor product of the modules $\Lambda^\bullet L_j^* \otimes \mathbb{C}$; both gradings and $F^{E/S}$ split accordingly ($\Gamma_{\mathrm{Cl}} = \prod_j ic(e^{2j-1})c(e^{2j})$ and the $F_j^{E/S}$ commute), and $2^{-m}\,\mathrm{tr}$ is the product of the factors $2^{-1}\,\mathrm{tr}$. Both relative characters are therefore products over j, and it suffices to treat one oriented plane with orthonormal coframe e^1, e^2 and curvature $R = \left(\begin{smallmatrix} 0 & \theta \\ -\theta & 0 \end{smallmatrix}\right)$, $\theta = 2\pi x$.

On the basis $1, e^1, e^2, e^{12}$, the Levi–Civita curvature acts by the derivation $e^1 \mapsto -\theta e^2$, $e^2 \mapsto \theta e^1$, and kills 1 and e^{12}. The spin term in the definition of $F^{E/S}$ is $\frac{1}{4}(-\theta c^1 c^2 + \theta c^2 c^1) = -\frac{\theta}{2} c^1 c^2$, where $c^j = c(e^j)$. With $c^j = \varepsilon(e^j) - \iota_{e_j}$, the operator $c^1 c^2$ maps

$$1 \mapsto e^{12}, \quad e^{12} \mapsto -1, \quad e^1 \mapsto e^2, \quad e^2 \mapsto -e^1.$$

Let A be the operator $1 \mapsto e^{12}$, $e^{12} \mapsto -1$, $e^1 \mapsto -e^2$, $e^2 \mapsto e^1$. Then $A^2 = -1$, $F^{E/S} = F^E + \frac{\theta}{2} c^1 c^2 = \frac{\theta}{2} A$, and

$$\exp\Big(\frac{iF^{E/S}}{2\pi}\Big) = \exp\Big(\frac{x}{2}\, iA\Big) = \cosh\frac{x}{2} + \sinh\frac{x}{2}\, iA,$$

because $(iA)^2 = 1$. Moreover $c^1 c^2 = A$ on even forms and $-A$ on odd forms, so $\Gamma_{\mathrm{par}}\Gamma_{\mathrm{Cl}} = iA$, while $\Gamma_{\mathrm{Cl}}\Gamma_{\mathrm{Cl}} = 1$. Since $\operatorname{tr}(iA) = 0$ and $\operatorname{tr}((iA)^2) = 4$,

$$\tfrac{1}{2}\operatorname{tr}\big(iA\, e^{xiA/2}\big) = 2\sinh\frac{x}{2}, \qquad \tfrac{1}{2}\operatorname{tr}\big(e^{xiA/2}\big) = 2\cosh\frac{x}{2}. \qquad \square$$

Theorem 6.2 (Chern–Gauss–Bonnet)**.** *If M is closed, oriented, and even-dimensional, then*

$$\chi(M) = \int_M e(TM) = \int_M \mathrm{Pf}\Big(\frac{R^{TM}}{2\pi}\Big).$$

Proof. Hodge theory identifies

$$\operatorname{ind} D^+_{\mathrm{dR}} = \sum_j (-1)^j \dim H^j(M;\mathbb{C}) = \chi(M).$$

Apply Theorem 5.5 to $\Lambda^\bullet T^*M_{\mathbb{C}}$ with the parity grading. By Chapter 1 and Lemma 6.1,

$$\widehat{A}(TM)\operatorname{ch}_{\mathrm{par}}(\Lambda^\bullet/S) = \prod_{j=1}^{m} \frac{x_j/2}{\sinh(x_j/2)}\, 2\sinh(x_j/2) = \prod_{j=1}^{m} x_j = \mathrm{Pf}\Big(\frac{R^{TM}}{2\pi}\Big).$$

This form already has degree n, and its class is $e(TM)$. $\square$

If $\dim M$ is odd, Poincaré duality gives $\chi(M) = 0$, and there is no top-degree Euler form.

For a closed oriented surface,

$$e(TM) = \frac{K}{2\pi}\, dA,$$

where K is Gaussian curvature. Hence

$$\int_M K\, dA = 2\pi\chi(M),$$

the classical Gauss–Bonnet formula.

Example 6.3 (Surface of revolution)**.** Parametrize the generating curve by arclength and write the induced metric as

$$g = ds^2 + f(s)^2 d\theta^2.$$

For the oriented orthonormal coframe $\vartheta^1 = ds$, $\vartheta^2 = f(s)d\theta$, Cartan's first structure equation

$$d\vartheta^i + \omega^i{}_j \wedge \vartheta^j = 0$$

gives

$$\omega^1{}_2 = -f'(s)d\theta, \qquad \omega^2{}_1 = f'(s)d\theta.$$

Since the structure group is $SO(2)$, the quadratic term in the curvature equation vanishes and

$$\Omega^1{}_2 = d\omega^1{}_2 = -f''(s)ds \wedge d\theta = -\frac{f''(s)}{f(s)}\vartheta^1 \wedge \vartheta^2.$$

Thus

$$K(s) = -\frac{f''(s)}{f(s)}.$$

This computation is frame-independent because a change of oriented orthonormal frame changes $\omega^1{}_2$ by an exact one-form and leaves $d\omega^1{}_2$ unchanged.

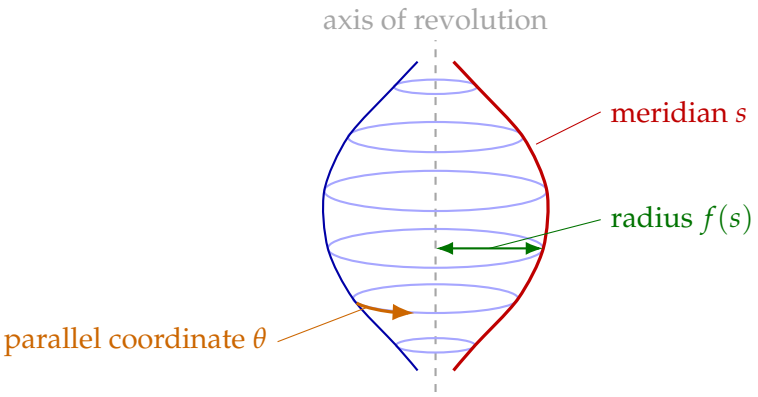


Figure 6.1: The arclength coordinate s follows a meridian, while θ runs around a parallel of radius $f(s)$. These orthogonal directions give $g = ds^2 + f(s)^2 d\theta^2$.

6.2 The signature operator

Let $\dim M = 4k$. On complex-valued p-forms define the chirality operator

$$\tau = i^{p(p-1)+2k} * .$$

Then

$$\tau^2 = 1, \qquad \tau D_{\mathrm{dR}} = -D_{\mathrm{dR}} \tau.$$

Let $\Omega^{\pm}$ be the ± 1-eigenspaces of τ. The signature operator is

$$D^{+}_{\mathrm{sign}} : \Omega^{+}(M) \longrightarrow \Omega^{-}(M), \qquad D^{+}_{\mathrm{sign}} = D_{\mathrm{dR}}|_{\Omega^{+}}.$$

Proposition 6.4.

$$\operatorname{ind} D^{+}_{\mathrm{sign}} = \operatorname{sign}(M),$$

the signature of the cup-product form on $H^{2k}(M;\mathbb{R})$.

Proof. The kernel of D_{dR} is the space of harmonic forms. For $p < 2k$, the involution τ exchanges $\mathcal{H}^p$ and $\mathcal{H}^{4k-p}$. On their direct sum it has matrix $\left(\begin{smallmatrix} 0 & U^{-1} \\ U & 0 \end{smallmatrix}\right)$ for an isomorphism U, so its $+1$ and -1 eigenspaces have equal dimension. In middle degree the phase $i^{p(p-1)+2k}$ equals one when $p = 2k$, so $\tau = *$. The positive and negative eigenspaces of $*$ have dimensions equal to the positive and negative indices of

$$([\alpha], [\beta]) \longmapsto \int_M \alpha \wedge \beta.$$

Their difference is the signature. □

Theorem 6.5 (Hirzebruch signature theorem)**.** *For a closed oriented* $4k$*-manifold,*

$$\boxed{\operatorname{sign}(M) = \int_M L(TM).}$$

Proof. Apply Theorem 5.5 to the Clifford module $(\Lambda^{\bullet} T^*M_{\mathbb{C}}, c, \tau)$, graded by τ rather than by parity. A check on monomials $e^{i_1} \wedge \cdots \wedge e^{i_p}$ shows that τ is the Clifford chirality $\Gamma_{\mathrm{Cl}} = i^{2k} c(e^1) \cdots c(e^{4k})$ [4, Chapter 3]; for instance both send 1 to $i^{2k}\, d\operatorname{vol}$. By Lemma 6.1, with $m = 2k$,

$$\widehat{A}(TM) \operatorname{ch}_{\tau}(\Lambda^{\bullet} T^*M_{\mathbb{C}}/S) = \prod_{j=1}^{2k} \frac{x_j/2}{\sinh(x_j/2)}\, 2\cosh(x_j/2) = \prod_{j=1}^{2k} \frac{x_j}{\tanh(x_j/2)}.$$

This is not the L-form in all degrees, but the two agree in degree $4k$. Indeed, with $g(y) = y/\tanh y$, $x/\tanh(x/2) = 2g(x/2)$. The degree-$4k$ part of a product of power series in $x_1, \dots, x_{2k}$ is its part of total polynomial degree $2k$, and the substitution $x_j \mapsto x_j/2$ multiplies that part by 2^{-2k}. Hence

$$\left[\prod_{j=1}^{2k} \frac{x_j}{\tanh(x_j/2)}\right]_{[4k]} = 2^{2k}\, 2^{-2k} \left[\prod_{j=1}^{2k} g(x_j)\right]_{[4k]} = \big[L(TM)\big]_{[4k]}. \tag{6.1}$$

The analytic index is the signature by Proposition 6.4. □

In dimension four,

$$L(TM)_{[4]} = \frac{1}{3} p_1(TM), \qquad \operatorname{sign}(M) = \frac{1}{3} \int_M p_1(TM).$$

Chapter 7

Complex geometry and Riemann–Roch

7.1 Almost complex and complex manifolds

An almost complex structure on a $2m$-manifold is an endomorphism $J : TM \to TM$ with $J^2 = -1$. Complexifying the tangent bundle gives

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M,$$

the $+i$ and $-i$ eigenspaces of J. The projections are

$$\Pi^{1,0} = \frac{1}{2}(I - iJ), \qquad \Pi^{0,1} = \frac{1}{2}(I + iJ).$$

For a real vector v,

$$v^{1,0} = \frac{1}{2}(v - iJv), \qquad v^{0,1} = \frac{1}{2}(v + iJv).$$

Dualizing and taking exterior powers gives

$$\Lambda^k T^*_{\mathbb{C}}M = \bigoplus_{p+q=k} \Lambda^{p,q} T^*M.$$

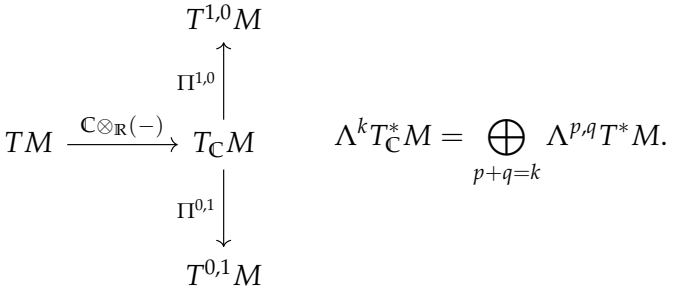


Figure 7.1: An almost complex structure splits the complexified tangent bundle and induces the (p, q)-decomposition of forms.

Proposition 7.1. *Every almost complex manifold is canonically oriented.*

Proof. Choose real vectors $v_1, \dots, v_m$ such that

$$v_1, Jv_1, \dots, v_m, Jv_m$$

is a basis and declare it positive. A second complex frame is related to the first by a matrix $A \in \mathrm{GL}(m, \mathbb{C})$. Viewed as a real matrix,

$$\det_{\mathbb{R}} A = |\det_{\mathbb{C}} A|^2 > 0.$$

Thus every complex change of frame preserves the declared orientation, which patches smoothly over M. □

The Nijenhuis tensor is

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

Theorem 7.2 (Newlander–Nirenberg)**.** *An almost complex structure is induced by holomorphic coordinate charts if and only if $N_J = 0$.*

Discussion. Newlander–Nirenberg is a nonlinear local existence theorem and is used here as a standard input. The necessity is elementary: coordinate vector fields of type $(1, 0)$ are closed under brackets, which forces $N_J = 0$. The converse solves a first-order nonlinear system for holomorphic coordinates and is not part of the index-theoretic proof.

Such a structure is called integrable, and M is then a complex manifold. On a complex manifold,

$$d = \partial + \bar{\partial},$$

with

$$\partial : \Omega^{p,q} \to \Omega^{p+1,q}, \qquad \bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1},$$

and $d^2 = 0$ implies

$$\partial^2 = 0, \qquad \bar{\partial}^2 = 0, \qquad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

On a nonintegrable almost complex manifold, d also has components of bidegrees $(2, -1)$ and $(-1, 2)$. In particular, the projected $\bar{\partial}$-operator does not generally square to zero. This distinction is essential: the canonical spin^c Dirac operator still exists, but there need not be a Dolbeault complex.

Question 7.3. *What exactly does integrability change?*

Answer. An almost complex structure always splits complexified forms by type and always determines a canonical spin^c symbol. Integrability is the extra condition that the $(0, 1)$-part of d squares to zero. Only then does $\bar{\partial}$ define cohomology and only then can the spin^c index be interpreted as a holomorphic Euler characteristic. □

Example 7.4 (Complex projective space)**.** The charts

$$U_j = \{[z_0 : \cdots : z_m] : z_j \neq 0\}, \qquad [z] \longmapsto \left(\frac{z_0}{z_j}, \dots, \widehat{\frac{z_j}{z_j}}, \dots, \frac{z_m}{z_j}\right)$$

have holomorphic transition functions and define the complex manifold $\mathbb{CP}^m$. Its induced almost complex structure is integrable by construction, and the Fubini–Study form makes it Kähler.

7.2 Almost Hermitian and almost Kähler geometry

An almost Hermitian structure is a pair (J, g) satisfying

$$g(JX, JY) = g(X, Y).$$

Its fundamental form

$$\omega(X, Y) = g(JX, Y)$$

is a nondegenerate real two-form. Compatible metrics always exist.

Question 7.5. *Does defining an almost complex structure require a metric?*

Answer. No. The datum is only the bundle endomorphism J with $J^2 = -I$, and it already determines the type decomposition and canonical orientation. A metric is introduced afterward. Averaging an arbitrary metric with its J-transform produces a compatible one, as the next proposition shows; there is no canonical choice in general. □

Proposition 7.6. *Every almost complex structure admits a compatible Riemannian metric.*

Proof. Start with any Riemannian metric g_0 and set

$$g(X, Y) = \frac{1}{2}\big(g_0(X, Y) + g_0(JX, JY)\big).$$

This is positive definite, and

$$g(JX, JY) = \frac{1}{2}\big(g_0(JX, JY) + g_0(J^2X, J^2Y)\big) = g(X, Y).$$

□

An almost Hermitian manifold is *almost Kähler* when

$$d\omega = 0.$$

It is *Kähler* when J is integrable and $d\omega = 0$. For an almost Hermitian structure,

$$\nabla^{TM} J = 0 \quad \Longleftrightarrow \quad N_J = 0 \text{ and } d\omega = 0.$$

Thus parallel J characterizes the Kähler condition, not the almost Kähler condition alone.

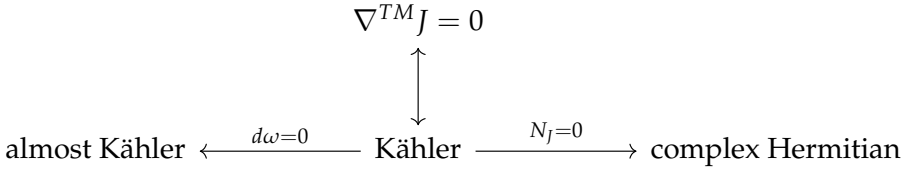


Figure 7.2: Kähler geometry is the simultaneous symplectic and integrable specialization of almost Hermitian geometry.

Discussion. The two lower conditions are independent in general. A symplectic manifold with a compatible almost complex structure is almost Kähler, but the chosen J need not be integrable. Conversely, a Hermitian metric on a complex manifold need not have closed fundamental form. Kähler geometry is the intersection of the symplectic and integrable conditions.

The Hermitian metric induces an inner product on (p,q)-forms. The complex-linear extension of the Hodge star has type

$$* : \Omega^{p,q}(M) \longrightarrow \Omega^{m-q,m-p}(M).$$

On a compact complex Hermitian manifold, define

$$\Box_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial.$$

This is elliptic, and Hodge theory gives

$$H^{p,q}_{\bar\partial}(M) \cong \ker\bigl(\Box_{\bar\partial} : \Omega^{p,q} \to \Omega^{p,q}\bigr).$$

Theorem 7.7 (Kähler identities)**.** *Let M be a Kähler manifold, let $L = \omega \wedge (-)$, and let $\Lambda = L^*$. Then*

$$[\Lambda, \partial] = i\bar\partial^*, \qquad [\Lambda, \bar\partial] = -i\partial^*.$$

Consequently,

$$\Box_\partial = \Box_{\bar\partial}, \qquad \Delta_d = 2\Box_{\bar\partial}.$$

Proof. Fix $p \in M$, choose a unitary Kähler-normal coframe $\{\zeta^j\}$, and apply both sides to an arbitrary test form α. At p, the connection coefficients vanish and $\nabla\omega = 0$, so

$$\partial\alpha = \sum_j \varepsilon(\zeta^j)\nabla_{Z_j}\alpha, \qquad \bar\partial\alpha = \sum_j \varepsilon(\bar\zeta^j)\nabla_{\bar Z_j}\alpha,$$

whereas

$$\partial^*\alpha = -\sum_j \iota_{Z_j}\nabla_{\bar Z_j}\alpha, \qquad \bar\partial^*\alpha = -\sum_j \iota_{\bar Z_j}\nabla_{Z_j}\alpha.$$

The algebraic relation

$$\iota_v\varepsilon(\alpha) + \varepsilon(\alpha)\iota_v = \alpha(v)$$

gives

$$[\Lambda, \varepsilon(\zeta^j)] = -i\iota_{\bar Z_j}, \qquad [\Lambda, \varepsilon(\bar\zeta^j)] = i\iota_{Z_j}.$$

Since $\nabla\Lambda = 0$, substitution in the preceding first-order formulas yields

$$[\Lambda, \partial]\alpha = i\bar\partial^*\alpha, \qquad [\Lambda, \bar\partial]\alpha = -i\partial^*\alpha.$$

The point and test form were arbitrary, so these are equalities of first-order differential operators.

The graded Jacobi identity and $\partial\bar\partial + \bar\partial\partial = 0$ give

$$\Box_{\bar\partial} = -i[\bar\partial, [\Lambda, \partial]] = i[\partial, [\Lambda, \bar\partial]] = \Box_\partial.$$

The same identities cancel the mixed terms in Δ_d, leaving $\Box_\partial + \Box_{\bar\partial} = 2\Box_{\bar\partial}$. □

It follows that

$$H^k(M;\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(M)$$

on every compact Kähler manifold: $\Delta_d = 2\Box_{\bar\partial}$ preserves bidegree, so the d-harmonic k-forms split into $\bar\partial$-harmonic (p,q)-forms. Without the Kähler hypothesis the identities and this decomposition fail in general; see [7, Chapter V].

7.3 The almost Dolbeault operator

On an almost complex manifold the type decomposition exists, but the exterior derivative has four components:

$$d = \mu + \partial + \bar\partial + \bar\mu,$$

of bidegrees

$$(2,-1), \qquad (1,0), \qquad (0,1), \qquad (-1,2).$$

The components μ and $\bar\mu$ are algebraic expressions in the Nijenhuis tensor and vanish exactly when J is integrable. Thus $\bar\partial$ exists by projection for every almost complex structure, but $\bar\partial^2$ need not vanish.

Nevertheless,

$$S = \Lambda^{0,\bullet} T^*M, \qquad S^\pm = \Lambda^{0,\mathrm{ev/odd}} T^*M$$

is the spinor bundle of the canonical spin^c structure. On a real covector ξ, Clifford multiplication is

$$c(\xi) = \sqrt{2}\left(\varepsilon(\xi^{0,1}) - \iota_{(\xi^\sharp)^{0,1}}\right), \qquad c(\xi)^2 = -|\xi|^2.$$

The compatible metric reduces the oriented frame bundle to $U(m)$ and thereby determines the canonical spin^c structure, whose determinant line is

$$K_M^{-1} = \det T^{1,0}M.$$

Choose a unitary connection A on K_M^{-1}. The Levi–Civita connection together with A lifts to a spin^c connection $\nabla^{S,A}$ on S. By construction it satisfies

$$[\nabla^{S,A}_X, c(\xi)] = c(\nabla^{LC}_X \xi),$$

and it is unitary and preserves the parity grading, so it is a Clifford connection in the sense of Definition 4.2 and defines a Dirac operator

$$D_J : C^\infty(S^+) \longrightarrow C^\infty(S^-).$$

An arbitrary Hermitian connection preserving J and g need not have this Clifford compatibility, because it need not induce the Levi–Civita connection on covectors. The spin^c construction is the required replacement.

The principal symbol of D_J is $i\,c(\xi)$, so it is elliptic. The operator $\sqrt{2}(\bar\partial + \bar\partial^*)$ has the same principal symbol: the symbols of $\bar\partial$ and $\bar\partial^*$ are exterior multiplication and contraction by $\xi^{0,1}$. Two first-order operators with the same principal symbol differ by a bundle endomorphism; here that zero-order term records the intrinsic torsion of the almost-Hermitian structure and the choice of A. As maps $H^1(S^+) \to L^2(S^-)$, their difference factors through the compact inclusion $H^1 \hookrightarrow L^2$. They therefore have the same index even when $\bar\partial^2 \neq 0$. In the Kähler case, choosing the Chern connection on K_M^{-1} gives $D_J = \sqrt{2}(\bar\partial + \bar\partial^*)$.

Let E be a Hermitian vector bundle with unitary connection ∇^E. The tensor-product connection on $S \otimes E$ is again a Clifford connection for $c \otimes 1$; write $D_{J,E}$ for its Dirac operator, the canonical spin^c Dirac operator twisted by E. For a unitary connection A on K_M^{-1}, write $c_1(A) = iF^A/2\pi$.

Theorem 7.8 (Spin^c index theorem)**.** *Let M^{2m} be compact and almost complex, with a compatible metric. Then*

$$\operatorname{ind} D^+_{J,E} = \int_M \left[e^{c_1(K_M^{-1})/2}\, \widehat{A}(TM)\, \operatorname{ch}(E) \right]_{[2m]}.$$

Proof. By Theorem 5.5 it suffices to show that the relative Chern character form of the Clifford module $S \otimes E$ is

$$\operatorname{ch}\big((S \otimes E)/S_{\text{spin}}\big) = e^{c_1(A)/2} \operatorname{ch}(E, \nabla^E).$$

The relative trace is intrinsic (Lemma 5.1), so this may be checked on a contractible open set U. There TM is spin, K_M has a square root, and the canonical spin^c structure satisfies

$$\Lambda^{0,\bullet} T^*M|_U \cong S_U \otimes K_U^{-1/2}$$

as graded Clifford modules, with the parity of the antiholomorphic degree corresponding to the Clifford chirality of the spinor bundle S_U; the spin^c connection is the tensor product of the spin connection and the connection on $K_U^{-1/2} = (K_U^{-1})^{1/2}$ induced by A [6, Appendix D]. Hence the twisting bundle is the ungraded bundle $W = K_U^{-1/2} \otimes E$, with curvature $F^{E/S} = \frac{1}{2}F^A \otimes 1 + 1 \otimes F^E$, and

$$\operatorname{tr}_W \exp\Big(\frac{iF^{E/S}}{2\pi}\Big) = \exp\Big(\frac{iF^A}{4\pi}\Big) \operatorname{tr} \exp\Big(\frac{iF^E}{2\pi}\Big) = e^{c_1(A)/2} \operatorname{ch}(E, \nabla^E).$$

For E trivial this is the identity $\operatorname{ch}(\Lambda^{0,\bullet}/S) = e^{c_1(K_M^{-1})/2}$. □

Lemma 7.9. *For a complex vector bundle V with underlying real oriented bundle $V_{\mathbb{R}}$,*

$$\operatorname{Td}(V) = e^{c_1(V)/2} \widehat{A}(V_{\mathbb{R}}).$$

Proof. By the splitting principle it suffices to check one Chern root x:

$$e^{x/2} \frac{x/2}{\sinh(x/2)} = \frac{x\, e^{x/2}}{e^{x/2} - e^{-x/2}} = \frac{x}{1 - e^{-x}}.$$

The roots of $V_{\mathbb{R}}$ in the sense of Chapter 1, computed with the complex structure of V, are the Chern roots of V, and $c_1(V) = \sum_j x_j$. □

Theorem 7.10 (Almost-complex Riemann–Roch)**.** *Let M^{2m} be compact almost complex and let E be a complex vector bundle. For the canonical spin^c Dirac operator twisted by E,*

$$\operatorname{ind} D^+_{J,E} = \int_M \Big[\operatorname{Td}(T^{1,0}M) \operatorname{ch}(E)\Big]_{[2m]}.$$

Proof. Since $c_1(K_M^{-1}) = c_1(\det T^{1,0}M) = c_1(T^{1,0}M)$ and $T^{1,0}M \cong (TM, J)$, Lemma 7.9 gives $e^{c_1(K_M^{-1})/2}\widehat{A}(TM) = \operatorname{Td}(T^{1,0}M)$. Substitute this in Theorem 7.8. □

7.4 Holomorphic vector bundles

In this section M is a complex manifold of complex dimension m, so that $\bar\partial^2 = 0$ on $\Omega^{0,\bullet}(M)$. A holomorphic structure on a complex vector bundle $E \to M$ is an operator

$$\bar\partial_E : \Omega^{0,q}(E) \longrightarrow \Omega^{0,q+1}(E)$$

satisfying the Leibniz rule and

$$\bar\partial_E^2 = 0.$$

It defines the Dolbeault complex

$$0 \longrightarrow \Omega^{0,0}(E) \xrightarrow{\bar\partial_E} \Omega^{0,1}(E) \longrightarrow \cdots \longrightarrow \Omega^{0,m}(E) \longrightarrow 0.$$

Its cohomology is denoted $H^{0,q}(M; E)$, or $H^q(M, \mathcal{O}(E))$.

Proposition 7.11. *The Dolbeault complex is elliptic.*

Proof. At a nonzero real covector ξ, the principal symbol of $\bar\partial_E$ is exterior multiplication by $i\xi^{0,1}$. The symbol complex is therefore the Koszul complex

$$\Lambda^{0,\bullet}T^*_xM \otimes E_x \xrightarrow{\xi^{0,1}\wedge} \Lambda^{0,\bullet+1}T^*_xM \otimes E_x.$$

Contraction by the dual of $\xi^{0,1}$, divided by $|\xi^{0,1}|^2$, is a contracting homotopy. Hence the symbol complex is exact.

Indeed, if

$$h_\xi = \frac{\iota_{(\xi^{0,1})^\sharp}}{|\xi^{0,1}|^2},$$

then exterior multiplication and contraction satisfy

$$h_\xi(\xi^{0,1}\wedge) + (\xi^{0,1}\wedge)h_\xi = I.$$

For a nonzero real covector ξ, its $(0,1)$-part cannot vanish: complex conjugation would also make its $(1,0)$-part vanish. Thus the homotopy is defined at every point of $T^*M \setminus 0$. □

Given a Hermitian metric on a holomorphic bundle, there is a unique unitary connection whose $(0,1)$-part is $\bar\partial_E$. It is the Chern connection. Its curvature has type $(1,1)$.

In a holomorphic frame with Hermitian metric matrix h, the Chern connection has

$$\nabla^{0,1} = \bar\partial, \qquad \nabla^{1,0} = \partial + h^{-1}\partial h,$$

with the placement of h^{-1} determined by the column-vector convention. Metric compatibility gives this formula, hence uniqueness. Since the $(0,1)$-part is integrable and the $(2,0)$-part cancels by the Maurer–Cartan identity, the curvature is of type $(1,1)$.

Now let M be compact, with a Hermitian metric, and give E a Hermitian metric. The Dolbeault–Dirac operator is

$$D_E = \sqrt{2}\,(\bar\partial_E + \bar\partial_E^*) \quad \text{on } \Omega^{0,\bullet}(M;E).$$

It acts from even to odd antiholomorphic degree and has the principal symbol of the canonical spin^c Dirac operator twisted by E. On a Kähler manifold the operators agree; in general they differ, for standard connection choices, by a zero-order torsion term and have the same index. Since $\bar\partial_E^2 = 0$, $D_E^2 = 2\Box_{\bar\partial_E}$ preserves the degree q, and Hodge theory for the elliptic Dolbeault complex gives $H^{0,q}(M;E) \cong \ker \Box_{\bar\partial_E}|_{\Omega^{0,q}}$, so that $\ker D_E = \bigoplus_q \ker \Box_{\bar\partial_E}|_{\Omega^{0,q}}$ and

$$\operatorname{ind} D_E^+ = \sum_{q=0}^{m} (-1)^q \dim H^{0,q}(M;E) =: \chi(M,E).$$

Figure 7.3 summarizes this identification.

7.5 Hirzebruch–Riemann–Roch

Recall that $K_M = \Lambda^{m,0}T^*M$ and that the canonical spin^c structure has determinant line K_M^{-1}.

Theorem 7.12 (Hirzebruch–Riemann–Roch)**.** *If M is a compact complex manifold of complex dimension m and E is a holomorphic vector bundle, then*

$$\boxed{\chi(M,E) = \int_M \Big[\operatorname{Td}(T^{1,0}M)\operatorname{ch}(E)\Big]_{[2m]}.}$$

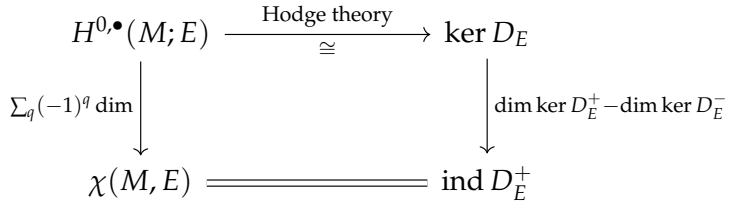


Figure 7.3: For an integrable complex structure, Dolbeault Hodge theory identifies the Dirac index with the holomorphic Euler characteristic. Here $H^{0,q}(M;E) \cong \ker D_E \cap \Omega^{0,q}(M;E)$.

Proof. Choose Hermitian metrics on M and E. The index of the Dolbeault–Dirac operator is $\chi(M,E)$ by Dolbeault Hodge theory (Figure 7.3). It has the same principal symbol as the canonical spin^c Dirac operator twisted by E; their difference is zero-order, so their indices agree. Theorem 7.10 computes the latter index. □

The characteristic number on the right is defined for every compact almost complex manifold and equals the index of its canonical spin^c Dirac operator; integrability is used only to identify that index with holomorphic sheaf cohomology, as in Question 7.3. For this circle of ideas see also [6] and [7].

7.6 Riemann surfaces

Let X be a compact connected Riemann surface of genus g, and let $L \to X$ be a holomorphic line bundle.

7.6.1 Divisors, meromorphic sections, and degree

A divisor is a finite formal sum

$$D = \sum_{p \in X} n_p [p], \qquad n_p \in \mathbb{Z},$$

and $\deg D = \sum_p n_p$. If f is meromorphic, its principal divisor is

$$(f) = \sum_p \operatorname{ord}_p(f)\,[p].$$

The argument principle on a compact Riemann surface gives $\deg(f) = 0$. Consequently degree descends from divisors to divisor classes.

Proposition 7.13 (Divisor–line-bundle correspondence)**.** *Every divisor determines a holomorphic line bundle $\mathcal{O}(D)$ with a distinguished meromorphic section s_D satisfying $(s_D) = D$. Conversely, a nonzero meromorphic section s of a holomorphic line bundle L determines a divisor (s) and an isomorphism*

$$L \cong \mathcal{O}((s)).$$

Two divisors define isomorphic line bundles precisely when they differ by a principal divisor.

Proof. Choose disks U_p around the support of D and a complementary open set U_0. On $U_p \cap U_0$, use the transition function $z_p^{-n_p}$. These functions form a holomorphic cocycle and define $\mathcal{O}(D)$; the local functions $z_p^{n_p}$ and 1 patch to a meromorphic section with divisor D. Conversely, local expressions $s = f_i e_i$ in holomorphic frames have orders that agree on overlaps, because the transition functions are nowhere zero. Dividing the local frame by the corresponding power of a coordinate identifies L with $\mathcal{O}((s))$. Replacing s by fs changes its divisor by (f), and the same local construction proves the last assertion. □

Dividing by the distinguished section s_D identifies sections of $\mathcal{O}(D)$ with meromorphic functions: a *holomorphic* section fs_D corresponds to a meromorphic function f that is allowed a pole of order at most n_p if $n_p > 0$, and is required to vanish to order at least $-n_p$ if $n_p < 0$. Equivalently,

$$H^0(X, \mathcal{O}(D)) \cong \{f \text{ meromorphic} : (f) + D \geq 0\} \cup \{0\}.$$

Proposition 7.14. *For a holomorphic line bundle on a compact Riemann surface,*

$$\deg L = \deg(s) = \int_X c_1(L)$$

for every nonzero meromorphic section s.

Proof. Choose disjoint coordinate disks around the zeros and poles of s, and use s as a nonvanishing holomorphic frame on their complement. On a disk about p, choose a nonvanishing holomorphic frame e_p and write

$$s = f_p e_p, \qquad f_p = z^{m_p} g_p, \qquad g_p(0) \neq 0.$$

The first Chern number is the sum of the winding numbers of these clutching functions. With the complex boundary orientation,

$$\int_X c_1(L) = \sum_p \frac{1}{2\pi i} \int_{|z|=\varepsilon} d \log f_p = \sum_p m_p.$$

The factor g_p has winding number zero. Thus the Chern number equals the divisor degree, independently of the chosen meromorphic section. □

In particular, the divisor of a meromorphic one-form represents the canonical bundle K_X. Since $c_1(K_X) = -c_1(T^{1,0}X)$, Gauss–Bonnet gives $\deg K_X = -(2 - 2g) = 2g - 2$.

The degree of L is therefore

$$\deg L = \int_X c_1(L).$$

Since only degree-two terms contribute,

$$\mathrm{ch}(L) = 1 + c_1(L), \qquad \mathrm{Td}(T^{1,0}X) = 1 + \frac{1}{2} c_1(T^{1,0}X).$$

Gauss–Bonnet gives

$$\int_X c_1(T^{1,0}X) = \chi(X) = 2 - 2g.$$

Hirzebruch–Riemann–Roch therefore becomes

$$\chi(X, L) = \deg L + 1 - g.$$

Theorem 7.15 (Serre duality for a curve)**.** *There is a natural perfect pairing*

$$H^1(X,L)\times H^0(X,K_X\otimes L^{-1})\longrightarrow \mathbb{C},\qquad ([\alpha],s)\longmapsto \int_X \alpha\wedge s.$$

Hence

$$H^1(X,L)\cong H^0(X,K_X\otimes L^{-1})^*.$$

Proof. First, the pairing is well defined on cohomology. If α is replaced by $\alpha+\bar\partial_L u$ and s is holomorphic, then

$$\int_X(\bar\partial_L u)\wedge s=\int_X \bar\partial(u\wedge s)=0$$

by Stokes' theorem.

Dolbeault theory identifies $H^1(X,L)$ with harmonic L-valued $(0,1)$-forms. Define the conjugate-linear Hermitian star

$$\bar *_L:\Omega^{0,1}(L)\longrightarrow \Omega^{1,0}(L^*)=\Gamma(K_X\otimes L^{-1})$$

by

$$\beta\wedge \bar *_L\alpha=\langle\beta,\alpha\rangle\, d\,\mathrm{vol}\qquad(\beta,\alpha\in\Omega^{0,1}(L)),$$

with the chosen complex orientation. Its formal-adjoint identity is

$$\bar\partial_{K_X\otimes L^{-1}}(\bar *_L\alpha)=-\bar *_L(\bar\partial_L^*\alpha).$$

Thus a harmonic α determines the holomorphic section $s_\alpha=\bar *_L\alpha$. Conversely, $\bar *_L^{-1}$ sends a holomorphic section to a top-degree $\bar\partial$-closed form annihilated by $\bar\partial_L^*$, hence to a harmonic form. Finally,

$$\int_X \alpha\wedge s_\alpha=\|\alpha\|_{L^2}^2.$$

Every nonzero harmonic class therefore has a nonzero pairing, and the inverse construction proves nondegeneracy in the other factor. This proves the perfect pairing and the stated duality. □

Combining Serre duality with the index formula gives the classical Riemann–Roch theorem.

Theorem 7.16 (Riemann–Roch for a line bundle)**.**

$$\boxed{h^0(X,L)-h^0(X,K_X\otimes L^{-1})=\deg L+1-g.}$$

Proof. Dolbeault theory gives

$$\chi(X,L)=h^0(X,L)-h^1(X,L).$$

Serre duality identifies

$$h^1(X,L)=h^0(X,K_X\otimes L^{-1}),$$

while Hirzebruch–Riemann–Roch in complex dimension one gives

$$\chi(X,L)=\int_X\left(c_1(L)+\frac{1}{2}c_1(T^{1,0}X)\right)=\deg L+1-g.$$

Combining the three equalities proves the formula. □

If a holomorphic line bundle has a nonzero holomorphic section, its degree is the number of zeros of that section counted with multiplicity and is therefore nonnegative. Thus, if $\deg L > 2g - 2$, then

$$\deg(K_X \otimes L^{-1}) < 0$$

and the second term vanishes:

$$h^0(X, L) = \deg L + 1 - g.$$

Discussion. The last vanishing uses only a simple divisor argument. A nonzero holomorphic section s has isolated zeros, and the local orders of vanishing sum to $\deg L$. Hence a negative-degree line bundle cannot have a nonzero holomorphic section. Since $\deg K_X = 2g - 2$, the bundle $K_X \otimes L^{-1}$ has negative degree under the displayed hypothesis.

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